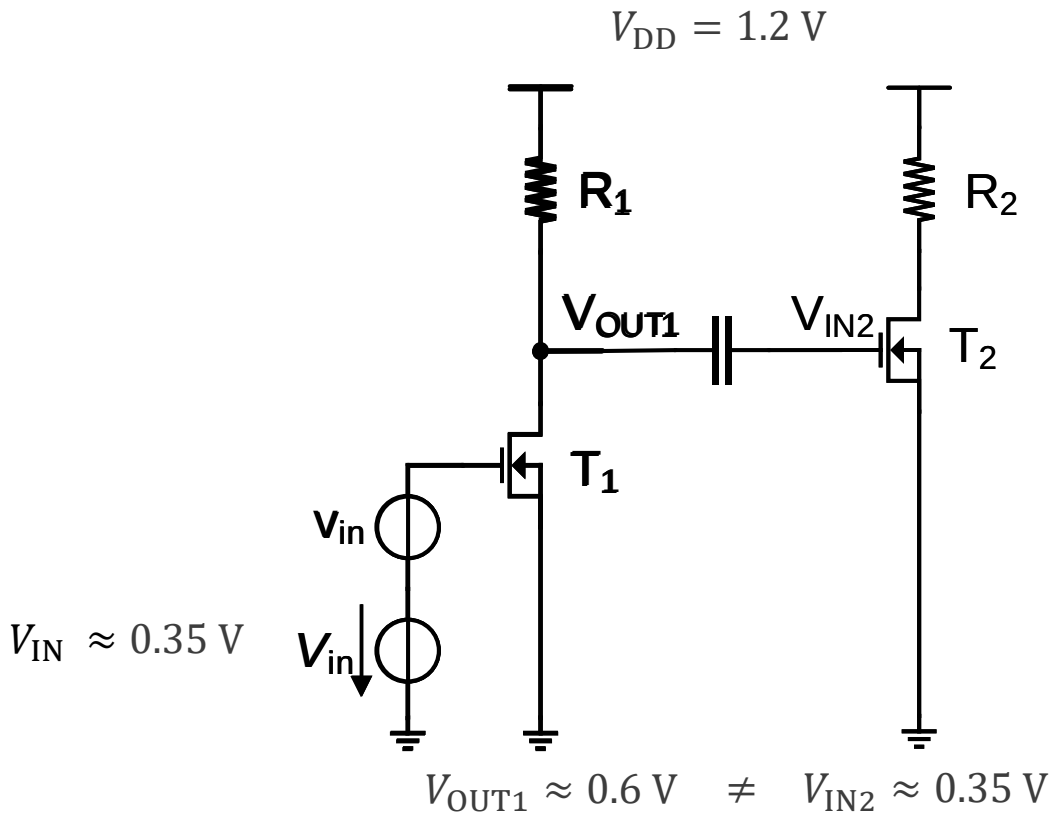
# Differential Stage

Winter Term 2025/26

# Overview

- **Motivation**
- Differential Stage
- Frequency Response

# Motivation for Self-Biasing I



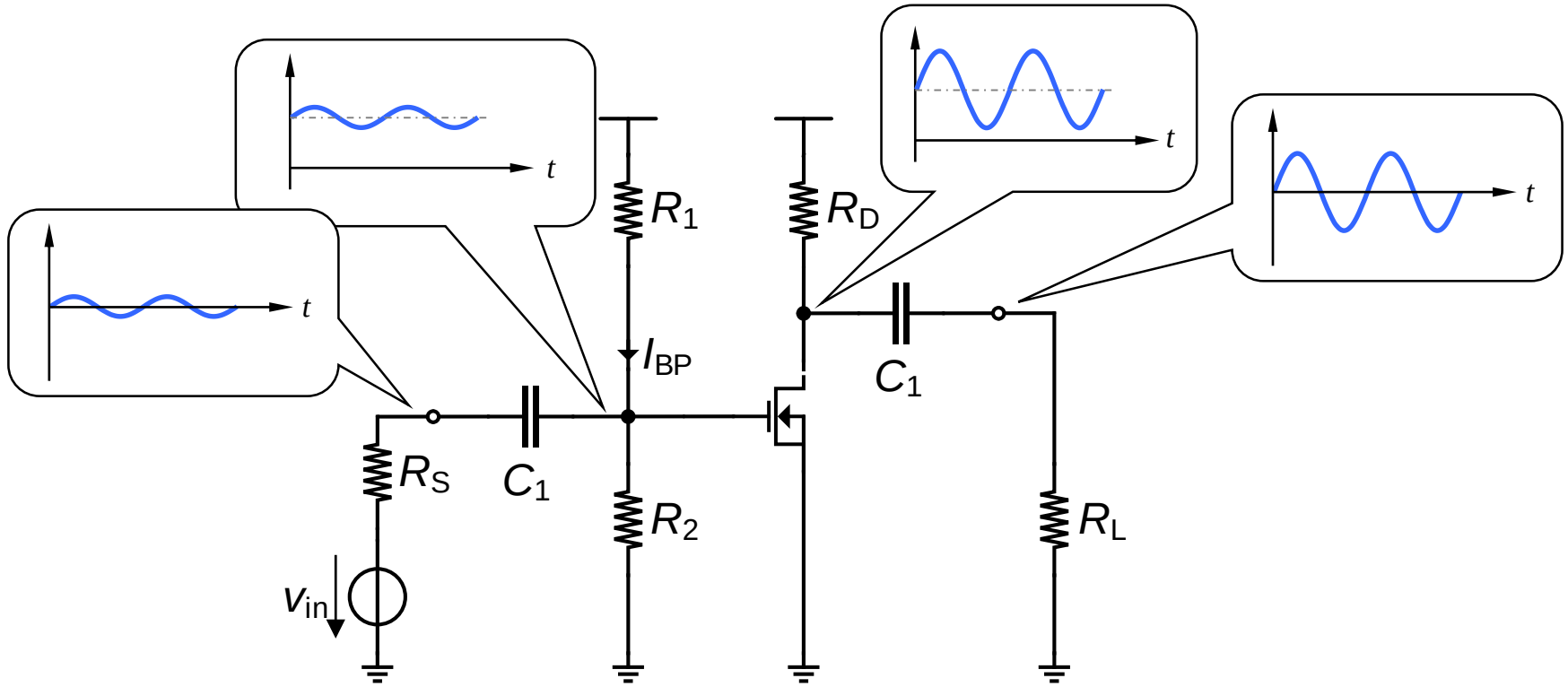

How to boost the gain?
→ cascade of amplifiers

Cascade of amplifiers for higher gain
→ coupling capacitor between stages for decoupling the output bias point of the previous stage and the input bias point of the next stage

# Defining Independent Bias Points – Coupling Capacitors

Coupling capacitors separate DC from AC signals.
DC currents are blocked from transmission!

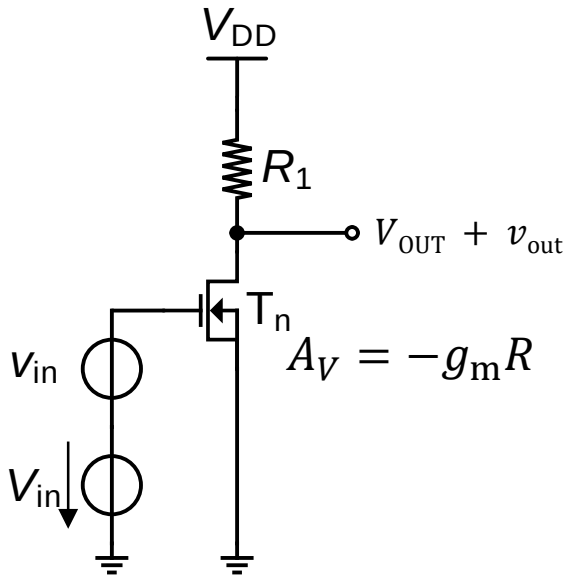
$C_1$ prevents a DC current flow into the signal source
(the resistance $R_S$ of the signal source would impact the voltage divider otherwise)

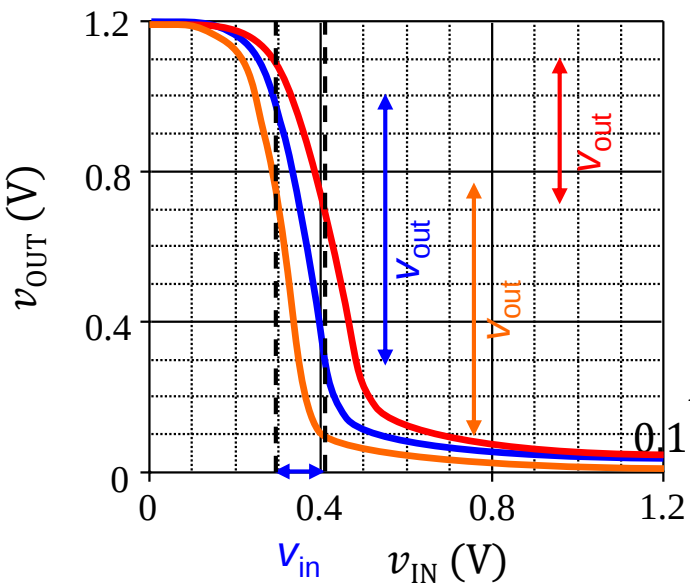


# Motivation for Self-Biasing II

Process, voltage and temperature (PVT) variations impact the bias point!

$$\beta_0, W/L, R, V_{GS}, V_{DD}, \text{temperature}$$




nominal values

$T_n$ corner fast, $R$ +20%
→ worst-case one

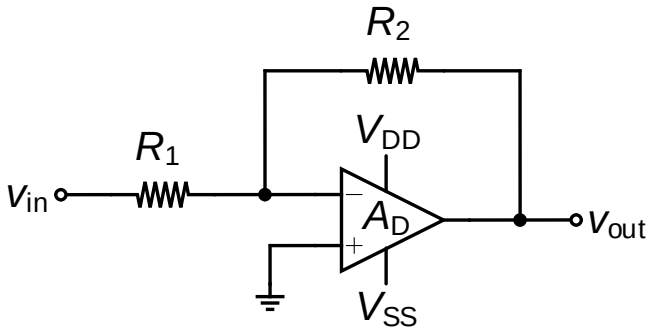
$T_n$ corner slow, $R$ -20%
→ worst-case zero

$R = 100\ \text{k}\Omega, V_{DD} = 1.2\ \text{V}$
$0.12\mu\text{m NMOS}, W/L = 10/10$

adaptable transfer characteristic → negative feedback around gain stage
→ differential amplifier

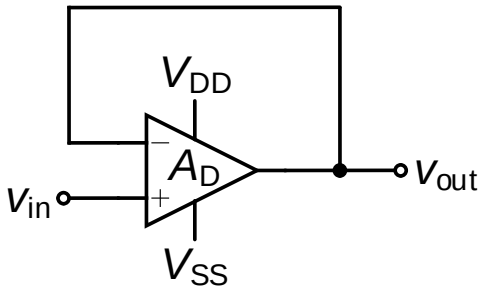
# Differential Amplifier

Circuits using differential amplifiers have no problems with biasing!




$$v_{\mathrm{OUT}} = -\frac{R_2}{R_1} v_{\mathrm{IN}} \rightarrow A_{\mathrm{v}} = -\frac{R_2}{R_1}$$

input: referred to zero (GND)
common mode: GND




$$v_{\mathrm{OUT}} = v_{\mathrm{IN}} \rightarrow \mathrm{A_v} = 1$$

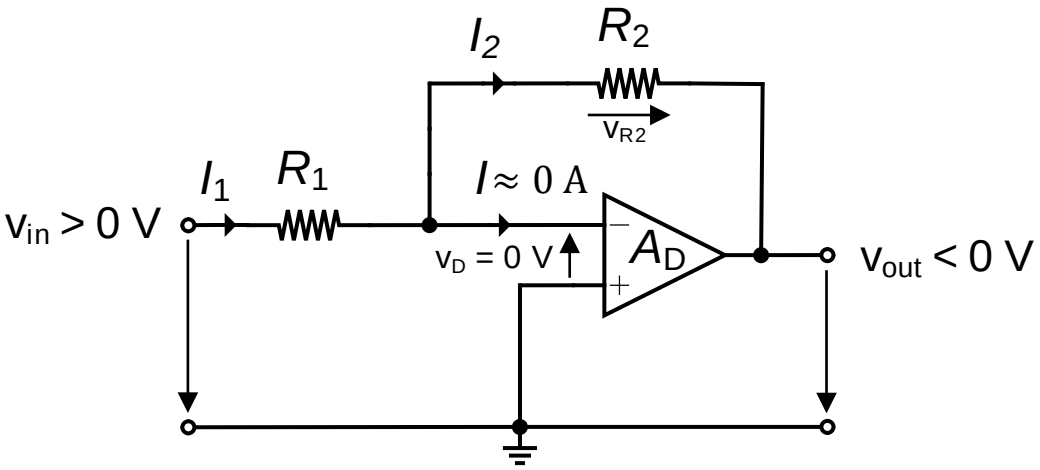
input: referred to zero (GND)
common mode determined by the input

supply:

$V_{\mathrm{DD}} = +0.6\ \mathrm{V}, V_{\mathrm{SS}} = -0.6\ \mathrm{V}$ (ground: 0 V) or

$V_{\mathrm{DD}} = +1.2\ \mathrm{V}, V_{\mathrm{SS}} = 0\ \mathrm{V}$ (ground: + 0.6 V → virtual GND, common mode)

# Inverting Amplifier




Two assumptions for the determination of $A_V$:

1. virtual short circuit: $v_D = 0\ \mathrm{V}$
   → inverting input at virtual ground
2. no current flow into the input of the opamp

due to 1: $I_1 = \frac{v_{R1}}{R_1} = \frac{v_{IN}}{R_1}$ $\quad I_2 = \frac{v_{R2}}{R_2} = -\frac{v_{OUT}}{R_2}$

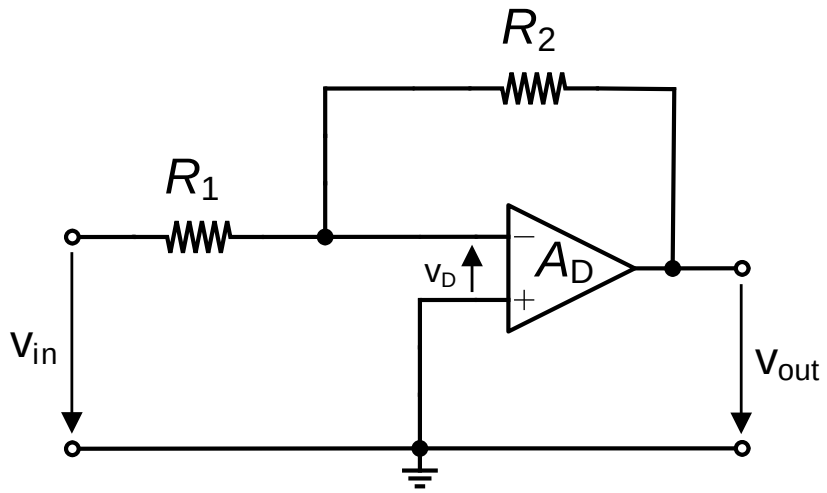
due to 2: $I_1 = I_2 \Rightarrow \frac{v_{IN}}{R_1} = -\frac{v_{OUT}}{R_2}$

The ideal opamp ($A_D \to \infty$) regulates its output voltage until $I_1 = I_2$.

$A_V = \frac{v_{OUT}}{v_{IN}}$ thus results in: $A_V = -\frac{R_2}{R_1}$

negative amplification → -180° phase shift
($v_{in} > 0\ \mathrm{V} \to v_{out} < 0\ \mathrm{V}$ and vice versa)

# Inverting Amplifier – Finite Differential Gain




For finite $A_D$, $A_V$ results in:

$$A_V = \frac{v_{OUT}}{v_{IN}} = -\frac{\frac{R_2}{R_1}}{1 + \frac{1}{A_D}\left(1 + \frac{R_2}{R_1}\right)}$$

example: $A_V = 100$ for $R_1 = 1\,\text{k}\Omega$ and $R_2 = 100\,k\Omega$.

How large is the error in percent of $A_V$ compared with the ideal value of $R_2 / R_1 = 100$ for different, finite open-loop gains $A_D$?

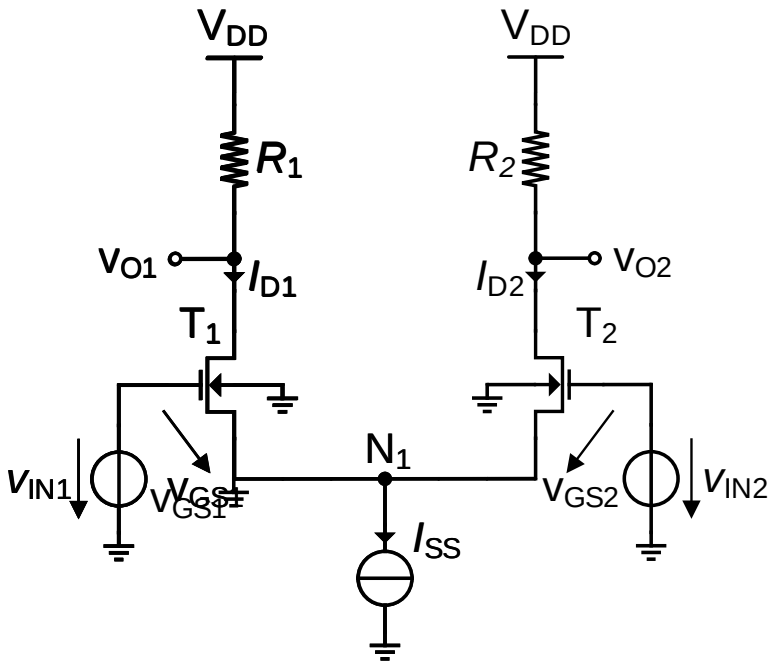
| $A_D$ | $v_{IN}$ | $v_{OUT}$ | $\lvert A_V \rvert$ | error (%) | $v_D$ |
|---|---|---|---|---|---|
| $10^5$ | 10 mV | | 99.90 | | 9.99 µV |
| $10^4$ | 10 mV | | 99.00 | | 99.0 µV |
| $10^3$ | 10 mV | | 90.83 | | 908 µV |

conclusion:
The error and $v_D$ are very small for large $A_D$!

# Overview

- Motivation
- **Differential Stage**
  - **Implementation & Large-Signal Behavior**
  - Input Common-Mode Range
  - Small-Signal Behavior
  - Load Elements
- Frequency Response

# Differential Stage




Self-biased stage through $I_{SS}$.
$v_{GS1}$ is defined by $W/L$ and $I_{SS}$

For $R_1 = R_2$ and $T_1 = T_2$
$(W_{1,2} = W, L_{1,2} = L, V_{th1,2} = V_{th}, \beta_{1,2} = \beta)$:

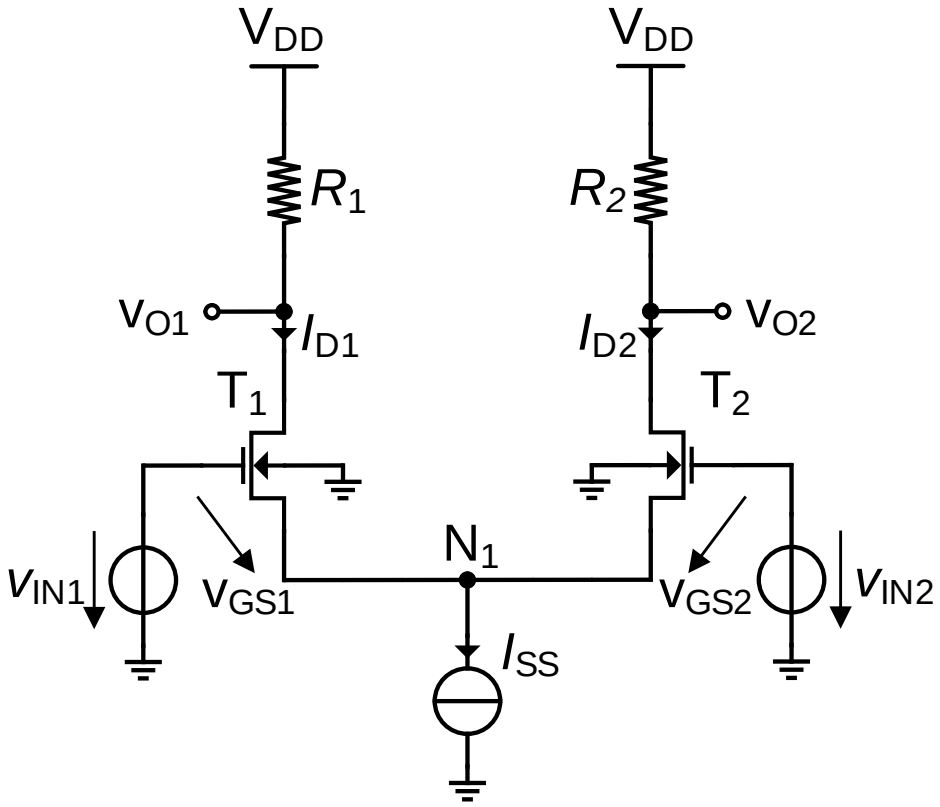
$$I_D = \frac{\beta_0}{2}\frac{W}{L}(v_{GS} - V_{th})^2 = \frac{\beta}{2}v_{GSeff}^2$$

always valid: $I_{D1} + I_{D2} = I_{SS}$

common-mode inputs $(v_{IN1} = v_{IN2})$: $I_{D1} = I_{D2} = \frac{I_{SS}}{2}$ $(v_{O1}, v_{O2}\ ?)$

differential-mode inputs $(\Delta v_{IN1} = -\Delta v_{IN2})$: $\Delta i_{D1} = -\Delta i_{D2}$ $(v_{O1}, v_{O2}\ ?)$

# Large-Signal Differential-Mode Behavior I




$$0\ \mathrm{V} = -v_{\mathrm{IN1}} + v_{\mathrm{GS1}} - v_{\mathrm{GS2}} + v_{\mathrm{IN2}}$$

$$\underbrace{v_{\mathrm{IN1}} - v_{\mathrm{IN2}}}_{\Delta v_{\mathrm{IN}}} = v_{\mathrm{GS1}} - v_{\mathrm{GS2}}$$

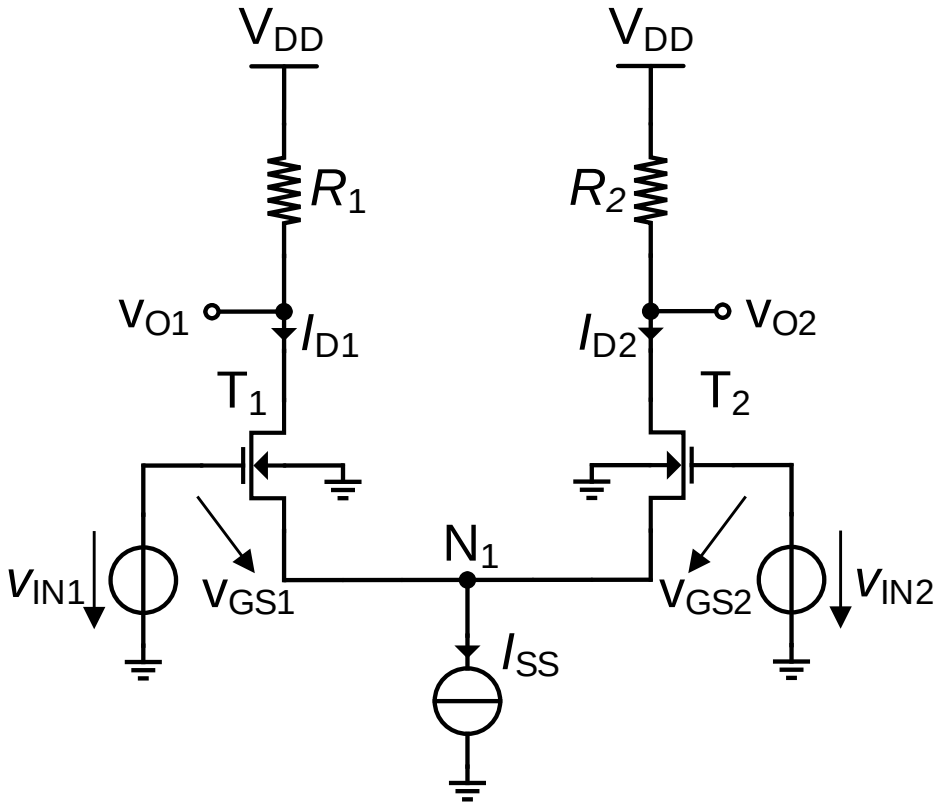
$$= v_{\mathrm{GSeff1}} + \cancel{V_{\mathrm{th1}}} - v_{\mathrm{GSeff2}} - \cancel{V_{\mathrm{th2}}}$$

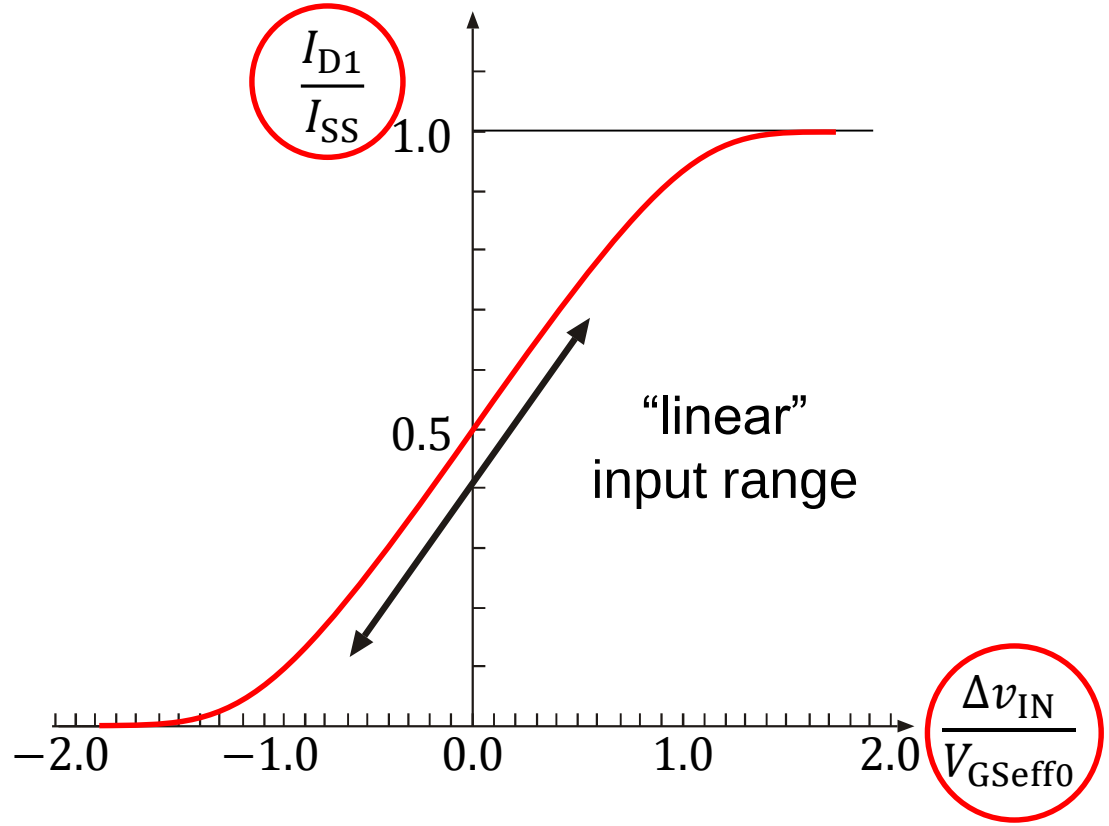
with $V_{\mathrm{th1}} = V_{\mathrm{th2}} = V_{\mathrm{th}}$ & $\beta_1 = \beta_2 = \beta$

$$I_{\mathrm{D}} = \frac{\beta}{2}(v_{\mathrm{GS}} - V_{\mathrm{th}})^2 \Rightarrow v_{\mathrm{GSeff}} = v_{\mathrm{GS}} - V_{\mathrm{th}} = \sqrt{\frac{2I_{\mathrm{D}}}{\beta}}$$

$$\Delta v_{\mathrm{IN}} = \sqrt{\frac{2I_{\mathrm{D1}}}{\beta}} - \sqrt{\frac{2I_{\mathrm{D2}}}{\beta}} \quad \text{with } I_{\mathrm{D1}} + I_{\mathrm{D2}} = I_{\mathrm{SS}}$$

# Large-Signal Differential-Mode Behavior II

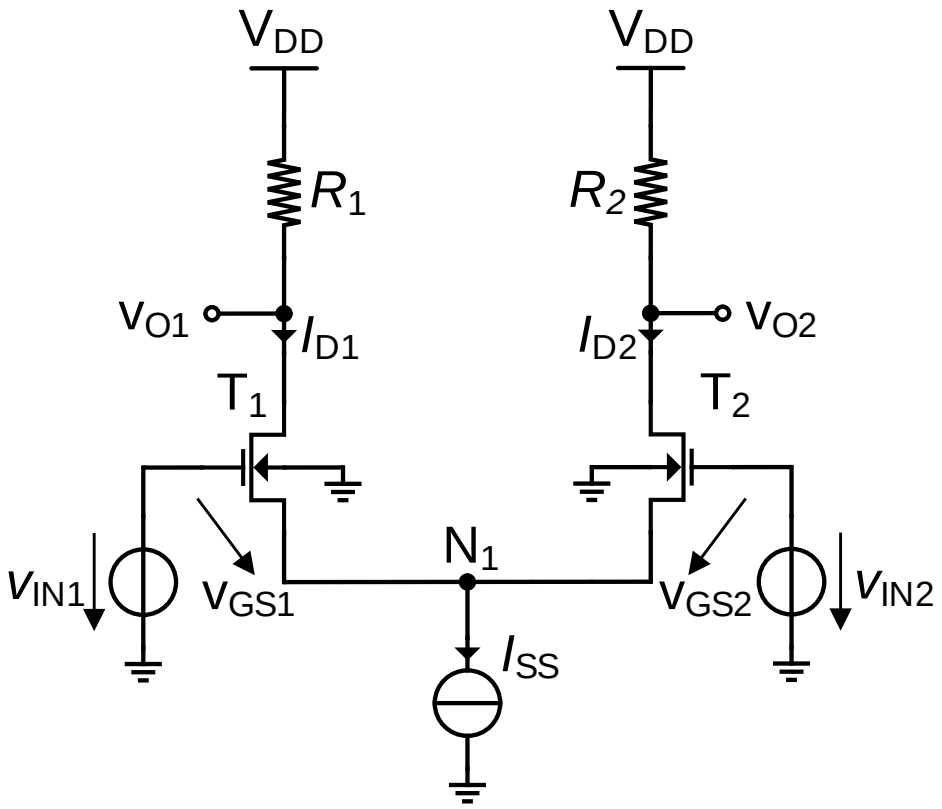


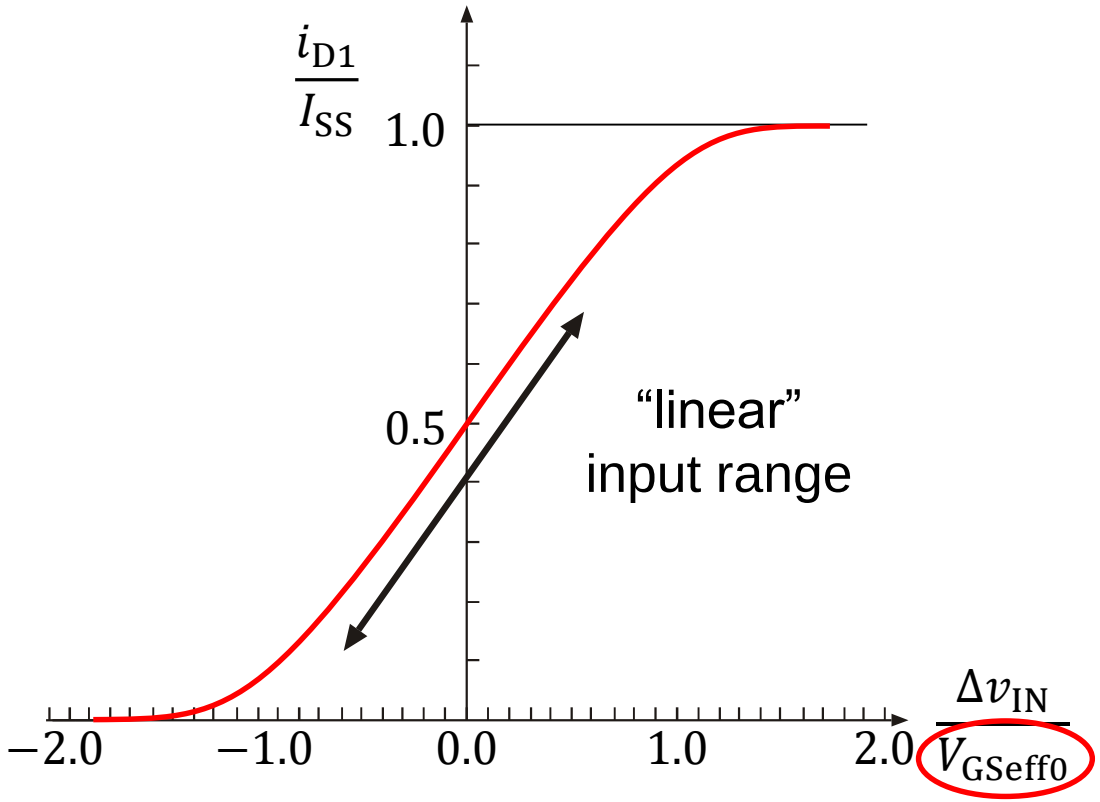




$$\Delta v_{\mathrm{IN}} = v_{\mathrm{GSeff1}} - v_{\mathrm{GSeff2}} = \sqrt{\frac{2I_{\mathrm{D1}}}{\beta}} - \sqrt{\frac{2I_{\mathrm{D2}}}{\beta}} \quad \text{with } I_{\mathrm{D1}} + I_{\mathrm{D2}} = I_{\mathrm{SS}}$$

$$\Rightarrow \quad I_{\mathrm{D1}} = \frac{I_{\mathrm{SS}}}{2}\left(1 + \Delta v_{\mathrm{IN}}\sqrt{\frac{\beta}{I_{\mathrm{SS}}} - \frac{\beta^2 \Delta v_{\mathrm{IN}}^2}{4I_{\mathrm{SS}}^2}}\right)$$

# Large-Signal Differential-Mode Behavior – $\Delta v_{\mathrm{IN}} = 0\ \mathrm{V}$




$$\Delta v_{\mathrm{IN}} = 0\ \mathrm{V} \Rightarrow I_{\mathrm{D1}} = I_{\mathrm{D2}} = \frac{I_{\mathrm{SS}}}{2}$$

$$\Rightarrow v_{\mathrm{GSeff1}} = v_{\mathrm{GSeff2}}$$

$$v_{\mathrm{GSeff}} = \sqrt{\frac{2I_{\mathrm{D}}}{\beta}} \Rightarrow \boxed{V_{\mathrm{GSeff0}} = \sqrt{\frac{I_{\mathrm{SS}}}{\beta}}}$$

# Large-Signal Differential-Mode Behavior III

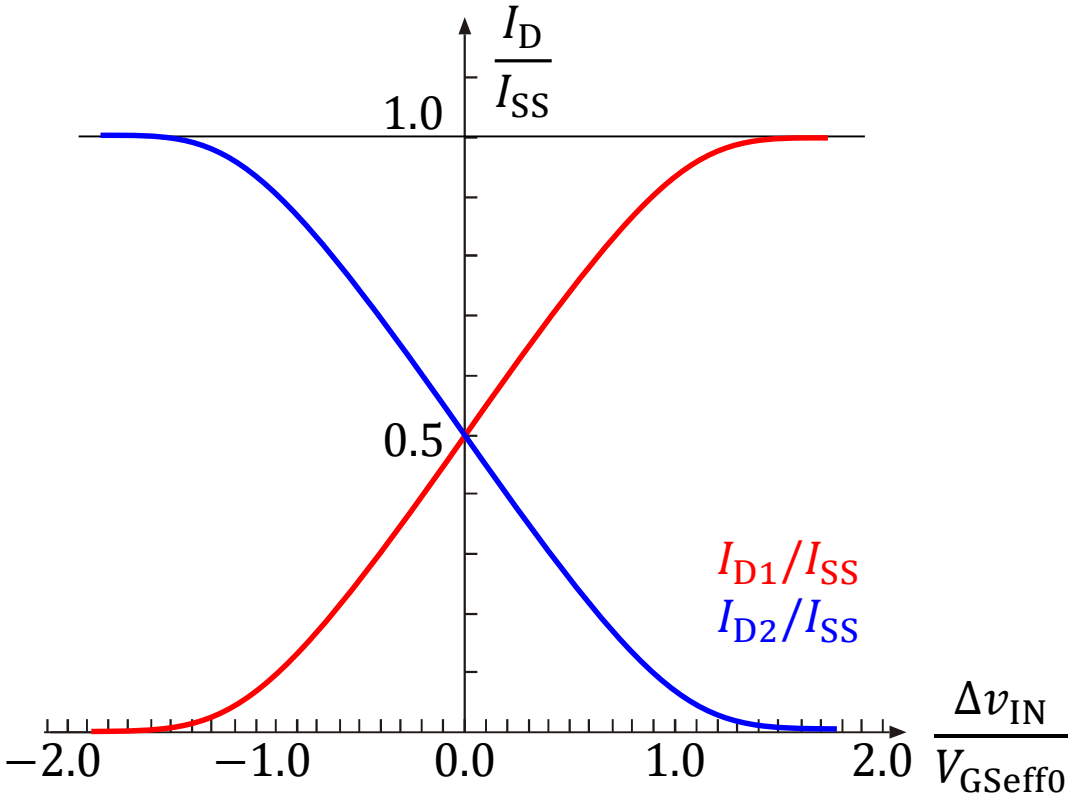
$$I_{D1} = \frac{I_{SS}}{2}\left(1 + \Delta v_{IN}\sqrt{\frac{\beta}{I_{SS}} - \frac{\beta^2 \Delta v_{IN}^2}{4I_{SS}^2}}\right)$$

with $V_{GSeff0} = \sqrt{\frac{I_{SS}}{\beta}}$

$$I_{D1} = \frac{I_{SS}}{2}\left(1 + \frac{\Delta v_{IN}}{V_{GSeff0}}\sqrt{1 - \frac{1}{4}\left(\frac{\Delta v_{IN}}{V_{GSeff0}}\right)^2}\right)$$

with $I_{D1} + I_{D2} = I_{SS}$

$$I_{D2} = \frac{I_{SS}}{2}\left(1 - \frac{\Delta v_{IN}}{V_{GSeff0}}\sqrt{1 - \frac{1}{4}\left(\frac{\Delta v_{IN}}{V_{GSeff0}}\right)^2}\right)$$



# Large-Signal Differential-Mode Behavior – $\Delta v_{\mathrm{IN}} \gg 0\ \mathrm{V}$

Increase $v_{\mathrm{IN1}}$ ($T_1$ draws more current) and decrease $v_{\mathrm{IN2}}$ ($T_2$ draws less current) until

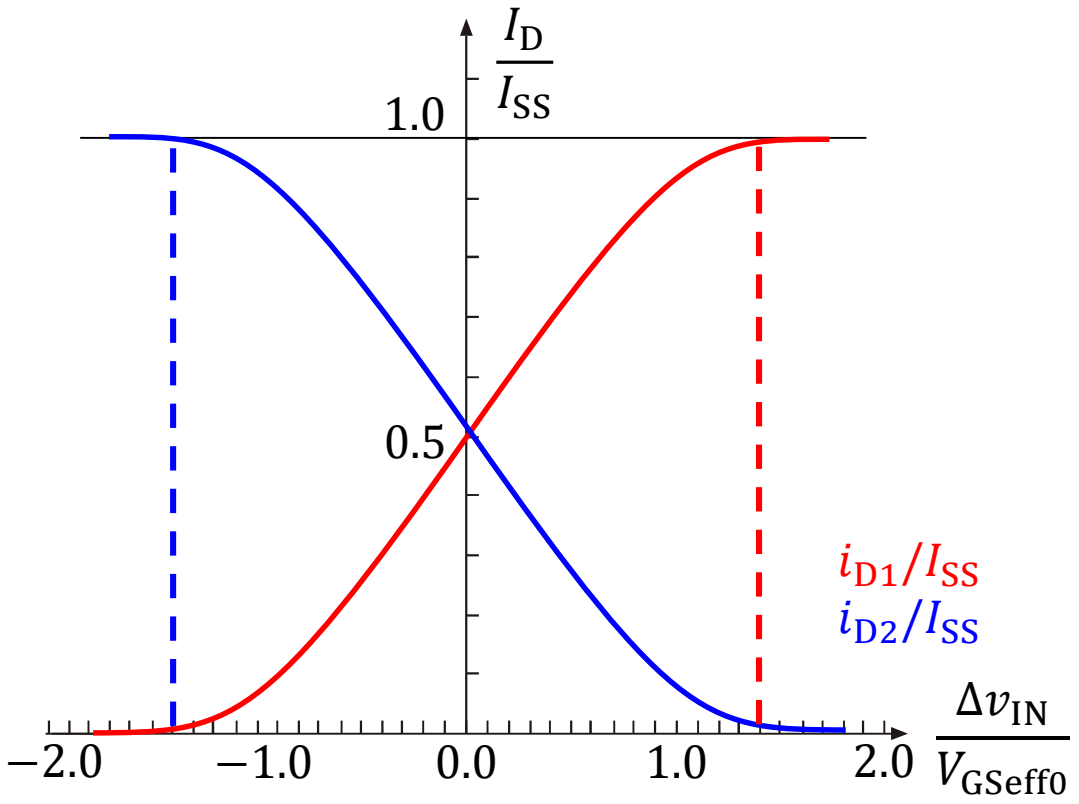
- $T_2$ is off:

  $v_{\mathrm{GS2}} \approx V_{\mathrm{th}} \rightarrow v_{\mathrm{GSeff2}} = 0\ \mathrm{V}$

- $T_1$ draws entire current $I_{\mathrm{SS}}$

$$v_{\mathrm{GSeff1max}} = \sqrt{\frac{2I_{\mathrm{SS}}}{\beta}} = \sqrt{2}V_{\mathrm{GSeff0}}$$

$$\begin{aligned}\Delta v_{\mathrm{INmax}} &= v_{\mathrm{GSeff1max}} - v_{\mathrm{GSeff2max}} \\ &= v_{\mathrm{GSeff1max}} \\ &= \sqrt{2}\, V_{\mathrm{GSeff0}}\end{aligned}$$



# Overview

- Motivation
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# Input Common-Mode Range – $V_{\mathrm{INmax}}$

Keep the input transistors in saturation!

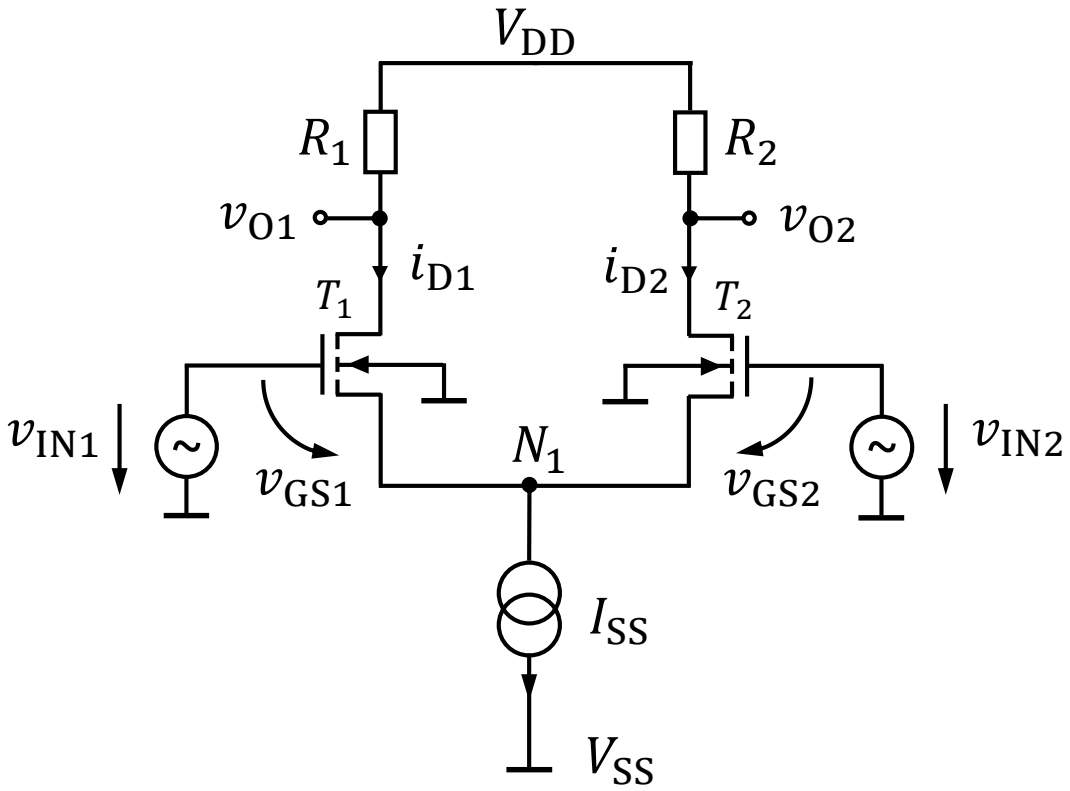
$$v_{\mathrm{DS}} \geq v_{\mathrm{GS}} - V_{\mathrm{th}}(v_{\mathrm{SB}})$$

$$\Leftrightarrow \quad v_{\mathrm{D}} \geq v_{\mathrm{G}} - V_{\mathrm{th}}(v_{\mathrm{SB}})$$

$$\Leftrightarrow \quad V_{\mathrm{Gmax}} = v_{\mathrm{D}} + V_{\mathrm{th}}(v_{\mathrm{SB}})$$

$$\Leftrightarrow \quad V_{\mathrm{INmax}} = V_{\mathrm{DD}} - Ri_{\mathrm{D1}} + V_{\mathrm{th}}(v_{\mathrm{SB}})$$

$$\boxed{V_{\mathrm{INmax}} = V_{\mathrm{DD}} - R\frac{I_{\mathrm{SS}}}{2} + V_{\mathrm{th}}(v_{\mathrm{SB}})}$$




example:

$I_{\mathrm{SS}} = 10\ \mu\mathrm{A} \quad R = 80\ \mathrm{k\Omega} \quad V_{\mathrm{DD}} = 1.2\mathrm{V}$

$V_{\mathrm{th0}} = 0.18\ \mathrm{V} \quad V_{\mathrm{SB}} = 0.8\ \mathrm{V} \quad \gamma = 0.1\ \sqrt{\mathrm{V}}$

$V_{\mathrm{th}} \approx V_{\mathrm{th0}} + \gamma\sqrt{V_{\mathrm{SB}}}$

$$V_{\mathrm{INmax}} = V_{\mathrm{DD}} - 0.4\ \mathrm{V} + 0.27\ \mathrm{V}$$
$$= V_{\mathrm{DD}} - 0.13\ \mathrm{V}$$

# Input Common-Mode Range – $V_{\mathrm{INmin}}$

Keep the current source operating, i.e., in saturation!

ideal current source: $V_{\mathrm{N1min}} = V_{\mathrm{Smin}} = V_{\mathrm{SS}}$

$$V_{\mathrm{INmin}} = V_{\mathrm{SS}} + \sqrt{\frac{I_{\mathrm{SS}}}{\beta}} + V_{\mathrm{th0}}$$

real current source (current mirror) $V_{\mathrm{N1min}} = V_{\mathrm{DSsat}}$

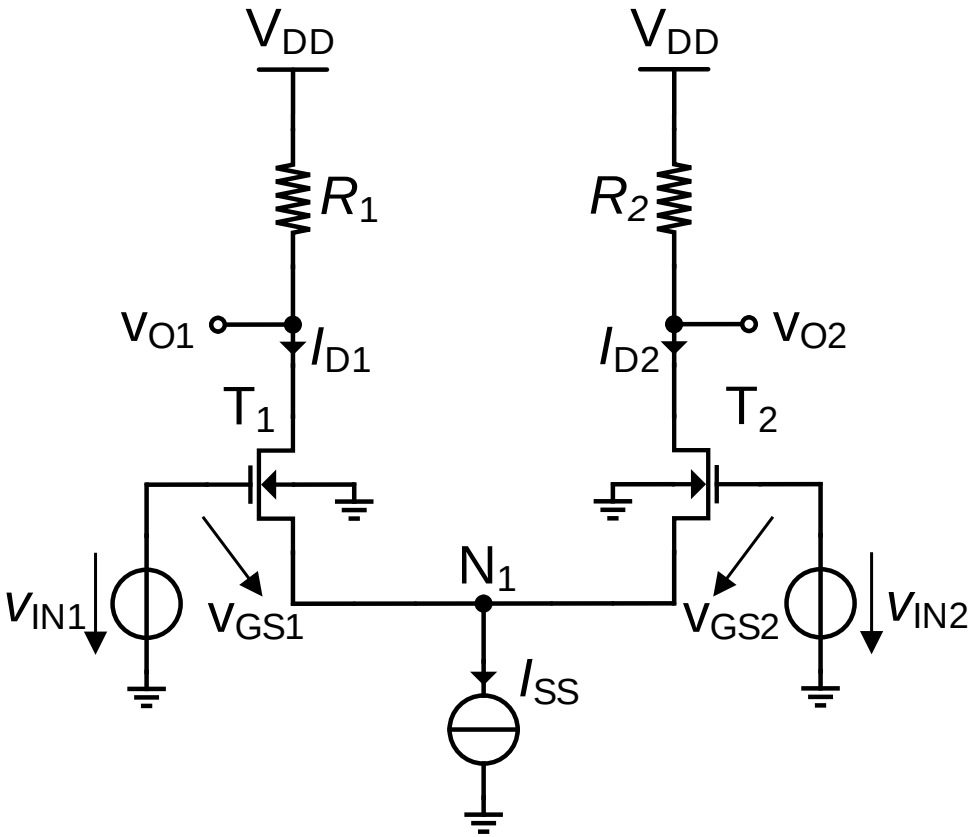
$$V_{\mathrm{INmin}} = V_{\mathrm{SS}} + V_{\mathrm{DSsat}} + \sqrt{\frac{I_{\mathrm{SS}}}{\beta}} + V_{\mathrm{th}}(v_{\mathrm{SB}})$$

example:

$I_{\mathrm{SS}} = 10\ \mu\mathrm{A} \quad \beta_0 = 360\ \frac{\mu\mathrm{A}}{V^2} \quad \frac{W}{L} = 10$

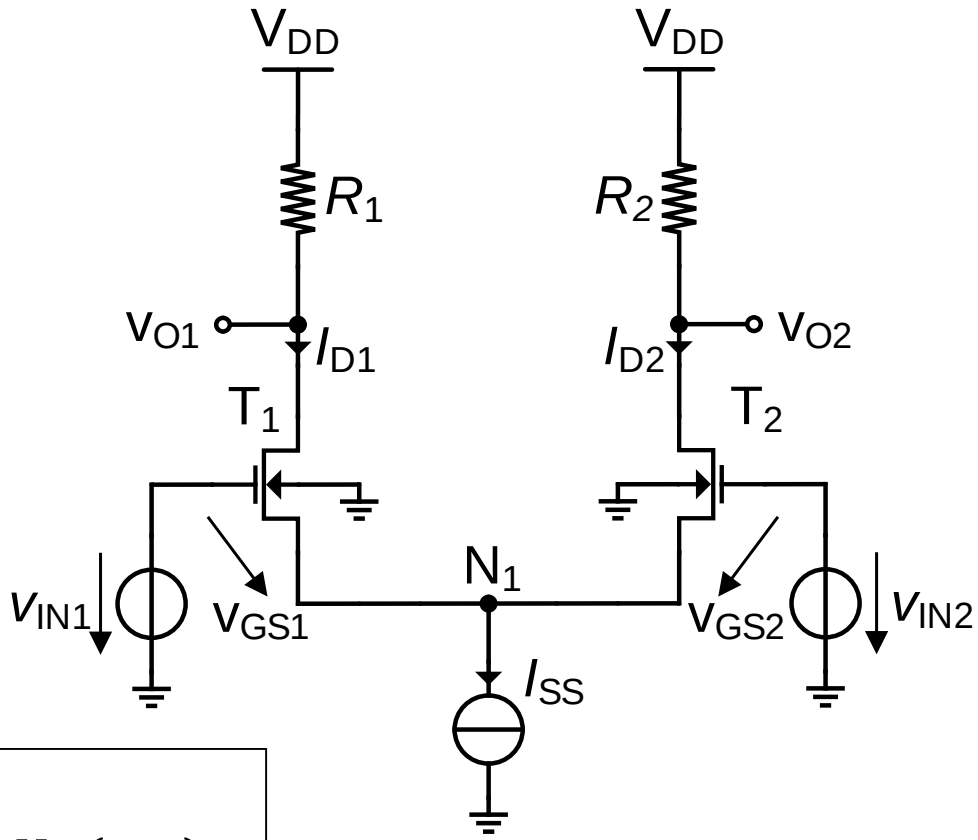
$V_{\mathrm{th0}} = 0.18\ \mathrm{V} \quad V_{\mathrm{SB}} = 0.8\ \mathrm{V} \quad \gamma = 0.1\ \sqrt{\mathrm{V}}$

$V_{\mathrm{th}} \approx V_{\mathrm{th0}} + \gamma\sqrt{V_{\mathrm{SB}}}$




$$V_{\mathrm{INmin}} = V_{\mathrm{SS}} + 0.053\ \mathrm{V} + 0.18\ \mathrm{V}$$
$$= V_{\mathrm{SS}} + 0.233\ \mathrm{V}$$

# Overall Input Common-Mode Range

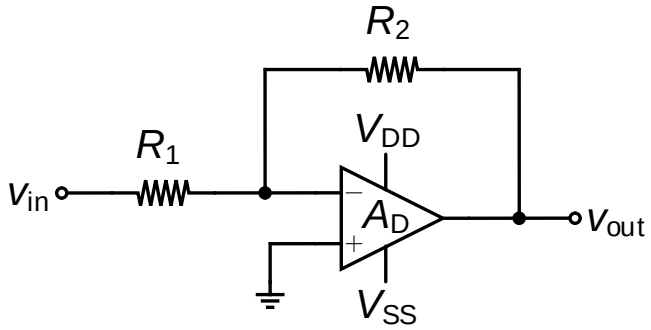



cascode: $2V_{\mathrm{DSsat}} + V_{\mathrm{th}}$

$$V_{\mathrm{SS}} + V_{\mathrm{DSsat}} + \sqrt{\frac{I_{\mathrm{SS}}}{\beta}} + V_{\mathrm{th}}(v_{\mathrm{SB}}) \leq V_{\mathrm{IN}} \leq V_{\mathrm{DD}} - R\frac{I_{\mathrm{SS}}}{2} + V_{\mathrm{th}}(v_{\mathrm{SB}})$$

# Common-Mode Range – Specification

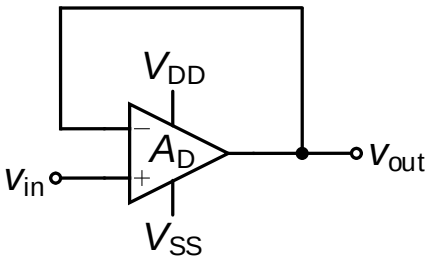
The requirement on the common-mode range is defined by the application!




$$v_{\mathrm{OUT}} = -\frac{R_2}{R_1} v_{\mathrm{IN}} \rightarrow A_{\mathrm{v}} = -\frac{R_2}{R_1}$$

amplifier has a fixed-common mode voltage defined by the non-inverting input (typically $\frac{V_{\mathrm{DD}}-V_{\mathrm{SS}}}{2}$)

→ opamp with low common-mode range is fine




$$v_{\mathrm{OUT}} = v_{\mathrm{IN}} \rightarrow \mathrm{A_v} = 1$$

common-mode voltage defined by the input signal (inverting input follows the non-inverting input)

→ opamp with high common-mode range required

# Overview

- Motivation
- **Differential Stage**
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  - Large-Signal Behavior
  - **Small-Signal Behavior**
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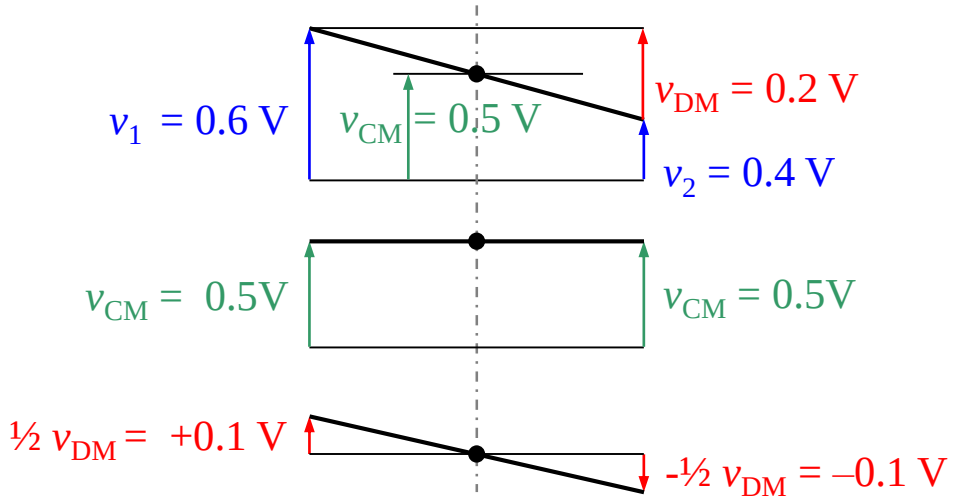
# Representation of Arbitrary Signals

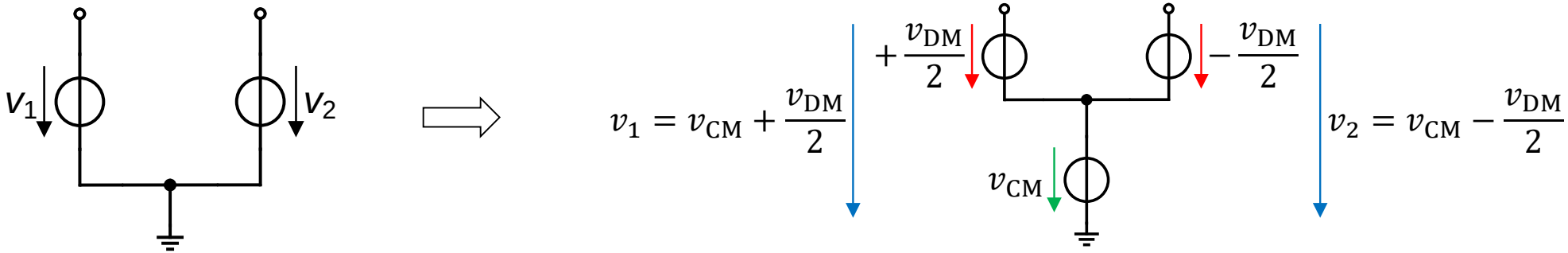
Two arbitrary signals can be represented by

- a common-mode (CM) and
- a differential-mode (DM) signal

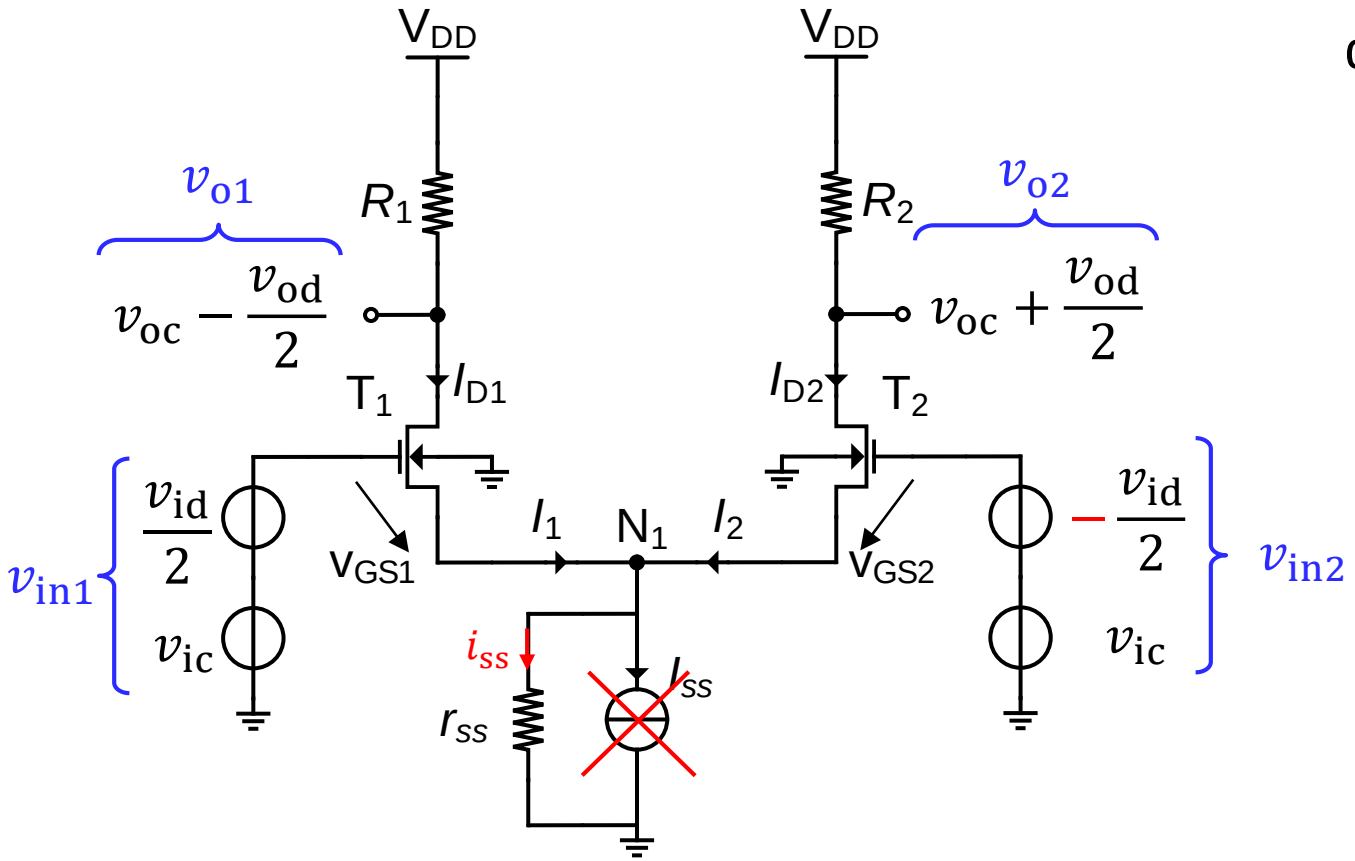
$$v_{\mathrm{CM}} = \frac{v_1 + v_2}{2} \qquad \text{(average)}$$

$$v_{\mathrm{DM}} = \mathrm{v}_1 - \mathrm{v}_2 \qquad \text{(difference)}$$



# Small-Signal Behavior / Amplification




differential- and common-mode voltages:

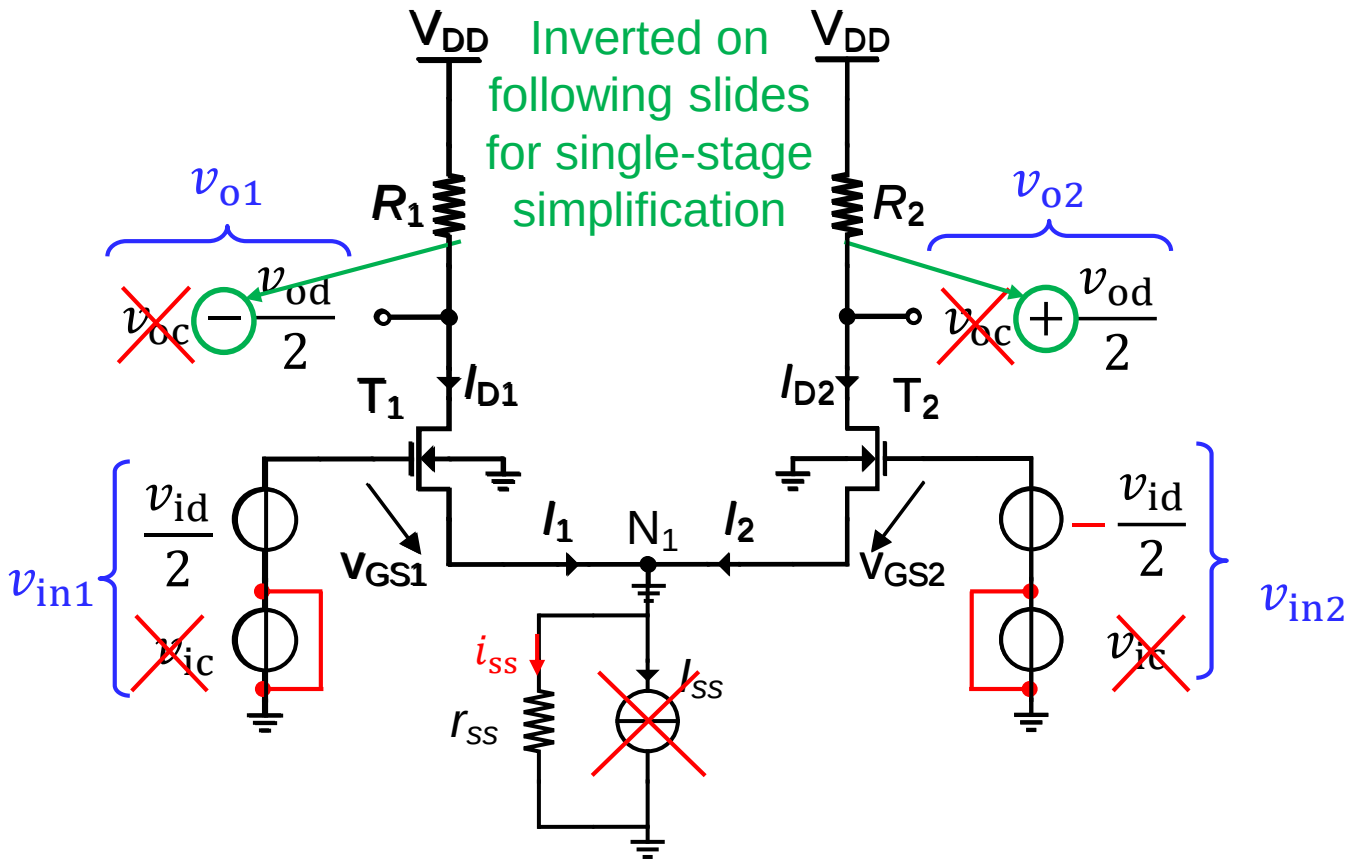
$$v_{in1} = v_{ic} + \frac{1}{2}v_{id} \qquad v_{o1} = v_{oc} - \frac{1}{2}v_{od}$$

$$v_{in2} = v_{ic} - \frac{1}{2}v_{id} \qquad v_{o2} = v_{oc} + \frac{1}{2}v_{od}$$

$$v_{id} = v_{in1} - v_{in2} \qquad v_{od} = v_{o1} - v_{o2}$$

$$v_{ic} = \frac{(v_{in1}+v_{in2})}{2} \qquad v_{oc} = \frac{(v_{o1}+v_{o2})}{2}$$

# Small-Signal Behavior / Amplification – Differential Mode




Since the principle of superposition can be applied to the linear small-signal model, we may analyze the differential- and common-mode small-signal amplification separately!

For the differential-mode small-amplification, it follows

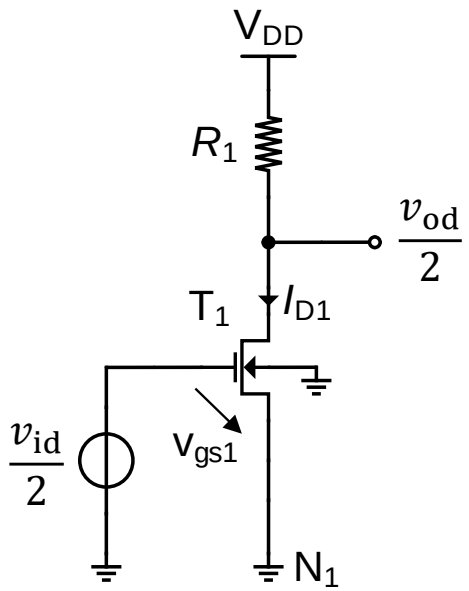
$$v_{ic} = 0\,\mathrm{V} \quad \Rightarrow \quad v_{oc} = 0\,\mathrm{V}$$

$$\frac{1}{2}v_{id} = \left|-\frac{1}{2}v_{id}\right| \Rightarrow i_1 = -i_2$$

$$\Rightarrow i_{ss} = 0\,\mathrm{A} \Rightarrow \Delta u_{N1} = 0$$

$\Rightarrow N_1$@ small-signal ground

# Differential Mode – Equivalent Small-Signal Model

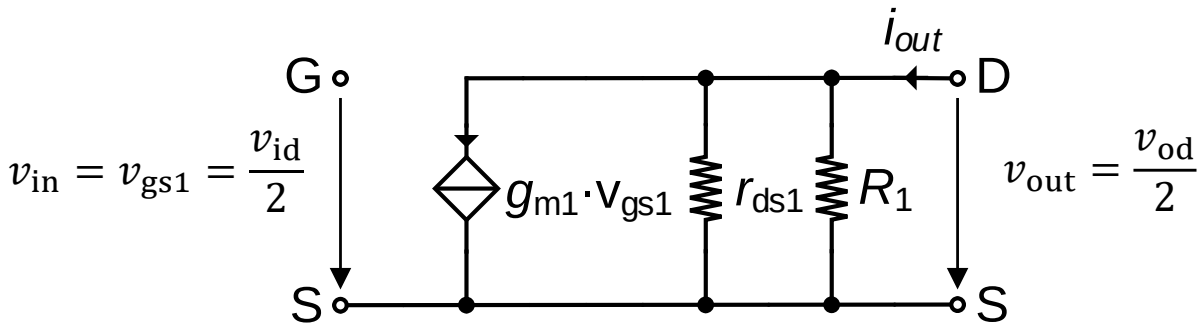



$$\frac{1}{2}v_{od} = A_V \frac{1}{2}v_{id}$$

$$A_V = -G_m r_{out}$$

$$= -g_{m1}(R_1 \parallel r_{ds1})$$

$$\boxed{A_{VDM} \approx -g_{m1}R_1}$$

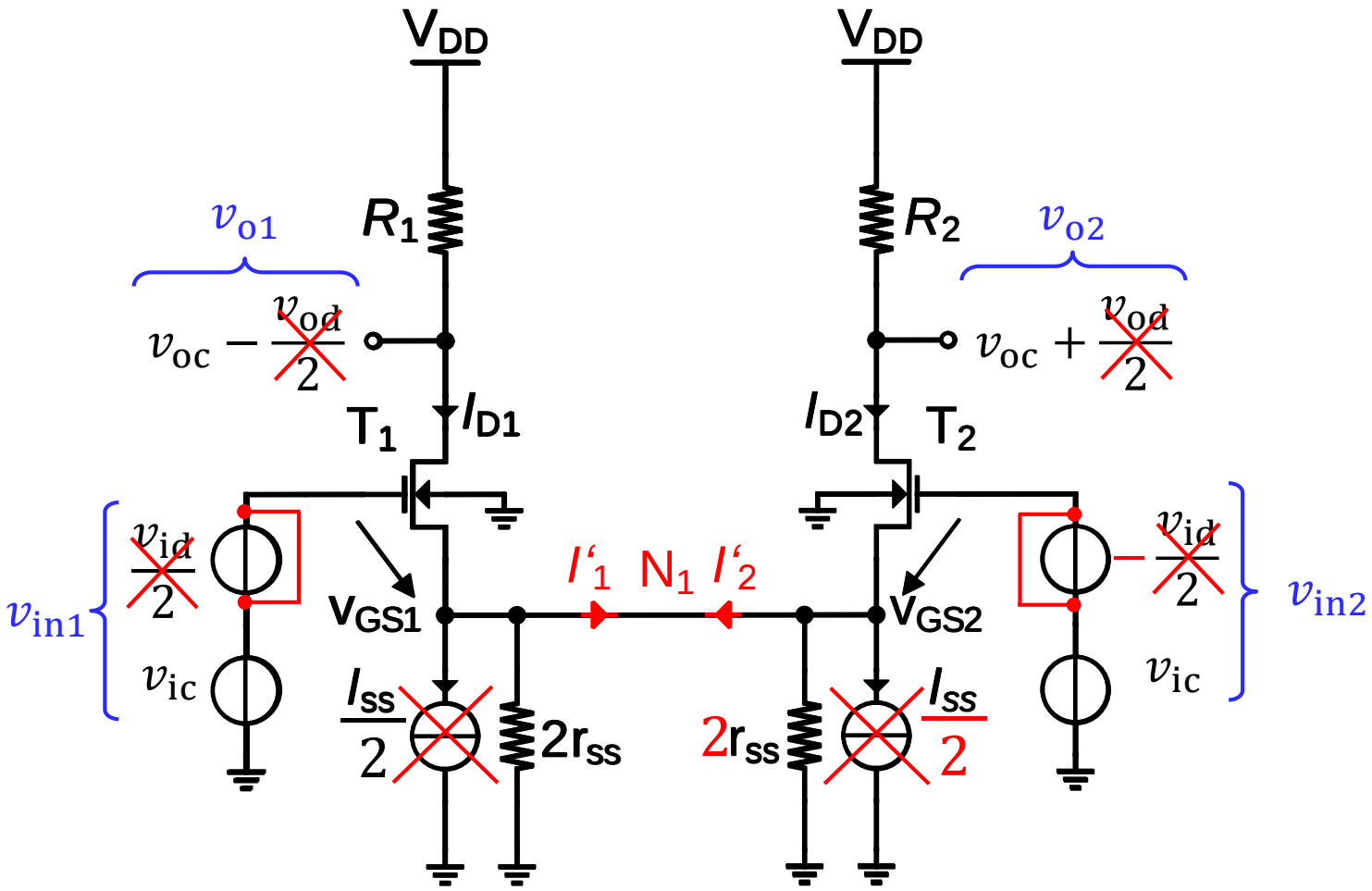



$$G_m = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out}=0\,V} = \frac{g_{m1}\frac{1}{2}v_{id}}{\frac{1}{2}v_{id}} = g_{m1}$$

$$r_{out} = \left.\frac{v_{out}}{i_{out}}\right|_{v_{in}=0\,V}$$

$$= r_{ds1} \parallel R_1 = \left(\frac{1}{R_1} + \frac{1}{r_{ds1}}\right)^{-1}$$

# Small-Signal Behavior / Amplification – Common Mode



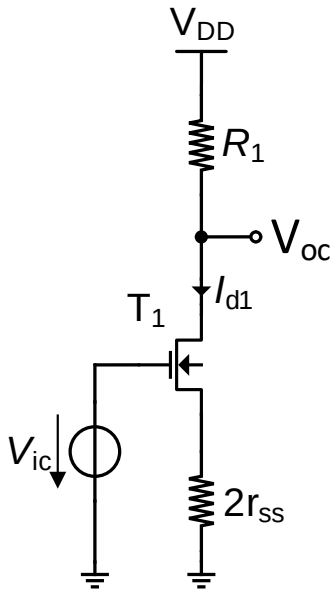

$$v_{ic} = v_{ic} \Rightarrow i_{d1} = i_{d2}$$

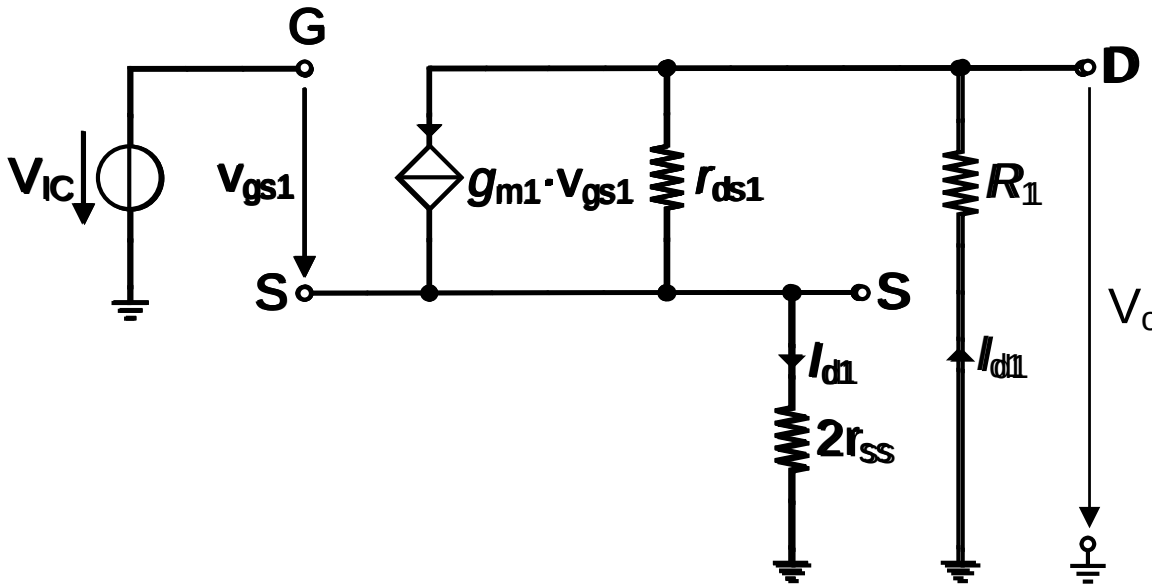
$i_1' = i_2' = 0\ \mathrm{A}$ due to symmetry

$\Rightarrow$ branch can be removed

$$I_{SS} = \frac{I_{SS}}{2} + \frac{I_{SS}}{2} \qquad r_{ss} = 2r_{ss} \parallel 2r_{ss}$$

# Common Mode – Equivalent Small-Signal Model




alternatives

$$A_{\mathrm{VCM}} = -G_{\mathrm{m}} r_{\mathrm{out}}$$

$$G_{\mathrm{m}} = \left.\frac{i_{\mathrm{out}}}{v_{\mathrm{ic}}}\right|_{v_{\mathrm{oc}} = 0\ \mathrm{V}} \qquad r_{\mathrm{out}} = \left.\frac{v_{\mathrm{oc}}}{i_{\mathrm{out}}}\right|_{v_{\mathrm{ic}} = 0\ \mathrm{V}}$$

$$v_{\mathrm{ic}} = v_{\mathrm{gs1}} + 2r_{\mathrm{ss}} i_{\mathrm{d1}}$$

$$v_{\mathrm{oc}} = -R_1 i_{\mathrm{d1}}$$

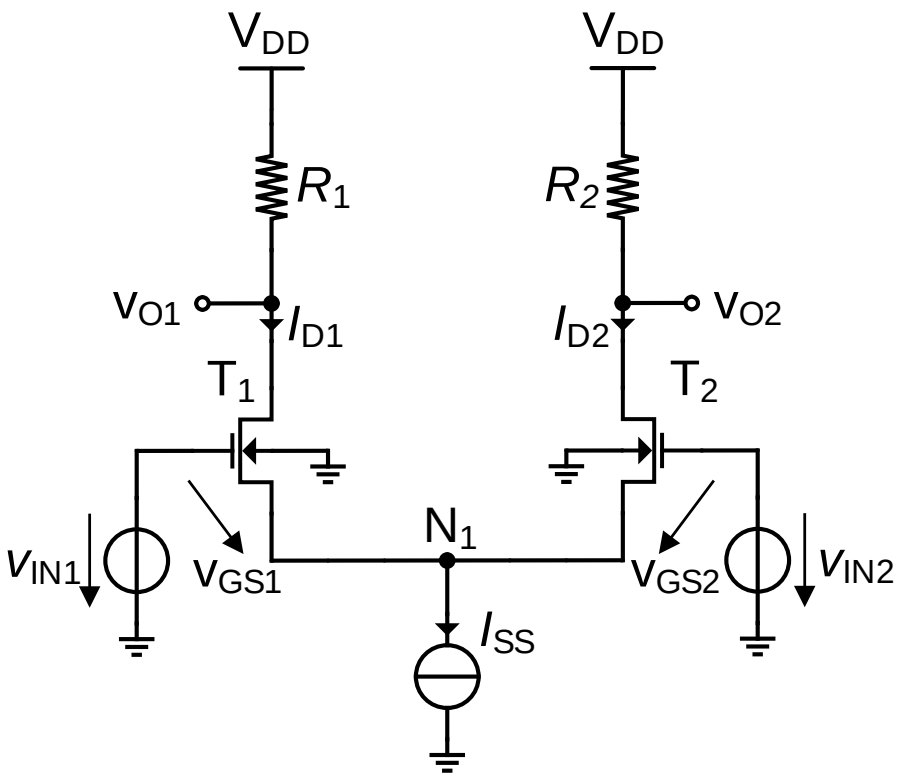
$$v_{\mathrm{oc}} = 2r_{\mathrm{ss}} i_{\mathrm{d1}} + r_{\mathrm{ds}}\left(i_{\mathrm{d1}} - g_{\mathrm{m}} v_{\mathrm{gs1}}\right)$$

$$A_{\mathrm{VCM}} = \frac{v_{\mathrm{oc}}}{v_{\mathrm{ic}}} = \frac{-g_{\mathrm{m1}}}{\frac{1}{r_{\mathrm{ds1}}} + \frac{1}{R_1} + 2\frac{r_{\mathrm{ss}}}{R_1}\left(\frac{1}{r_{\mathrm{ds1}}} + g_{\mathrm{m1}}\right)} = \frac{-g_{\mathrm{m1}}}{\frac{1}{R_1}\left(1 + 2g_{\mathrm{m1}} r_{\mathrm{ss}}\right)}$$

$$\boxed{A_{\mathrm{VCM}} \approx -\frac{R_1}{2r_{\mathrm{ss}}}}$$

with $r_{\mathrm{ds1}} \gg R_1$, $g_{\mathrm{m1}} \gg g_{\mathrm{ds1}} = \frac{1}{r_{\mathrm{Ds1}}}$ and $2g_{\mathrm{m1}} r_{\mathrm{ss}} \gg 1$

# Common Mode Rejection Ratio – CMRR




$$A_{VDM} \approx -g_m R$$

$$A_{VCM} \approx -\frac{R}{2r_{ss}}$$

$$CMRR = \frac{A_{VDM}}{A_{VCM}} = 2g_m r_{ss}$$

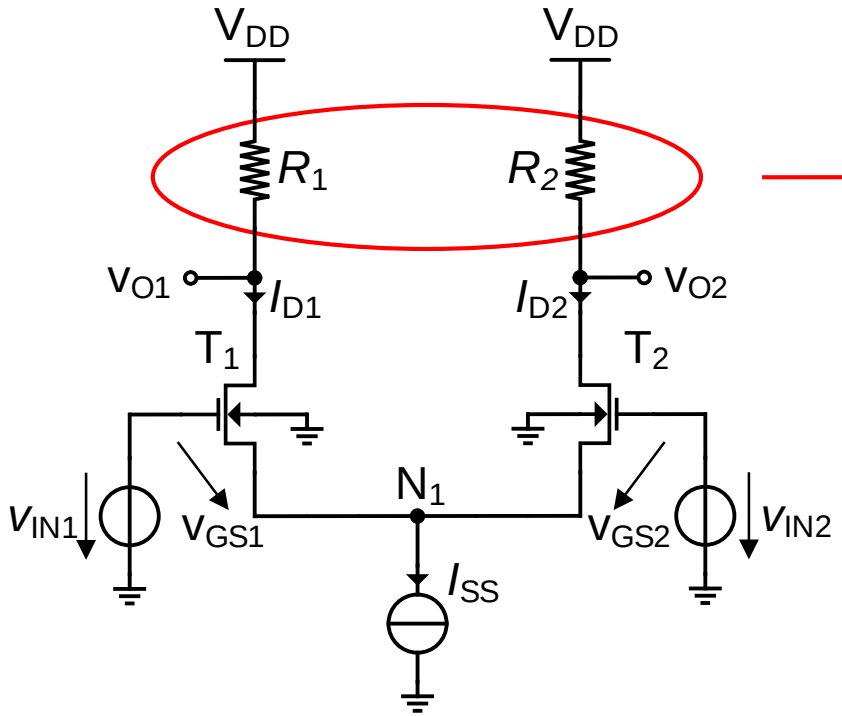
The CMRR is a measure for how much higher the differential-mode signal is amplified relative to the common-mode signal.

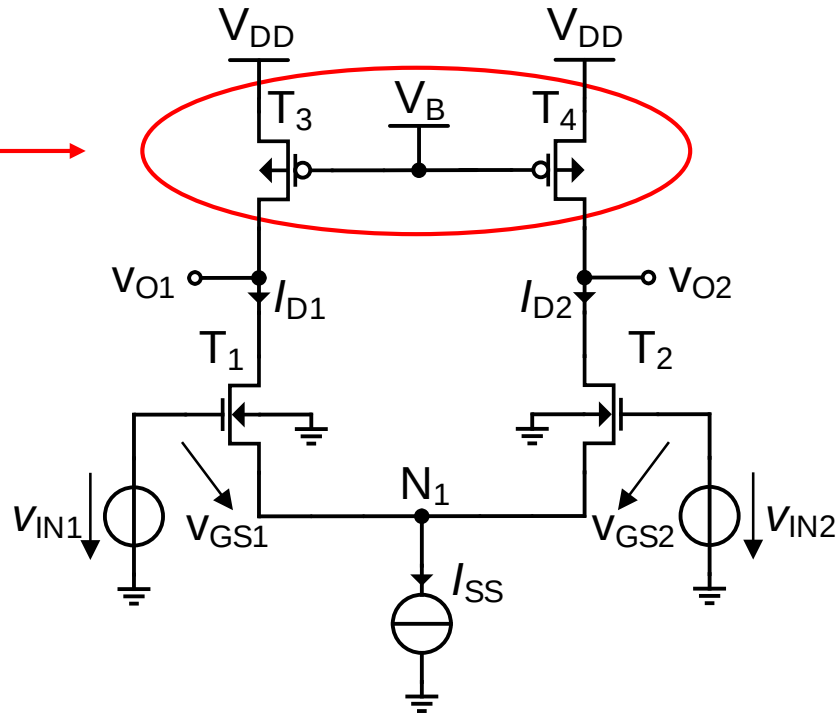
The higher the small-signal output resistance of the current source, the better the CMRR!
current mirror ⟷ cascode current mirror

# Overview

- Motivation
- **Differential Stage**
  - Implementation & Input Common-Mode Range
  - Large-Signal Behavior
  - Small-Signal Behavior
  - **Load Elements**
- Frequency Response

# Current Source Load




resistive load:

trade-off bias point (small $R$) vs. amplification (large $R$)

current source:

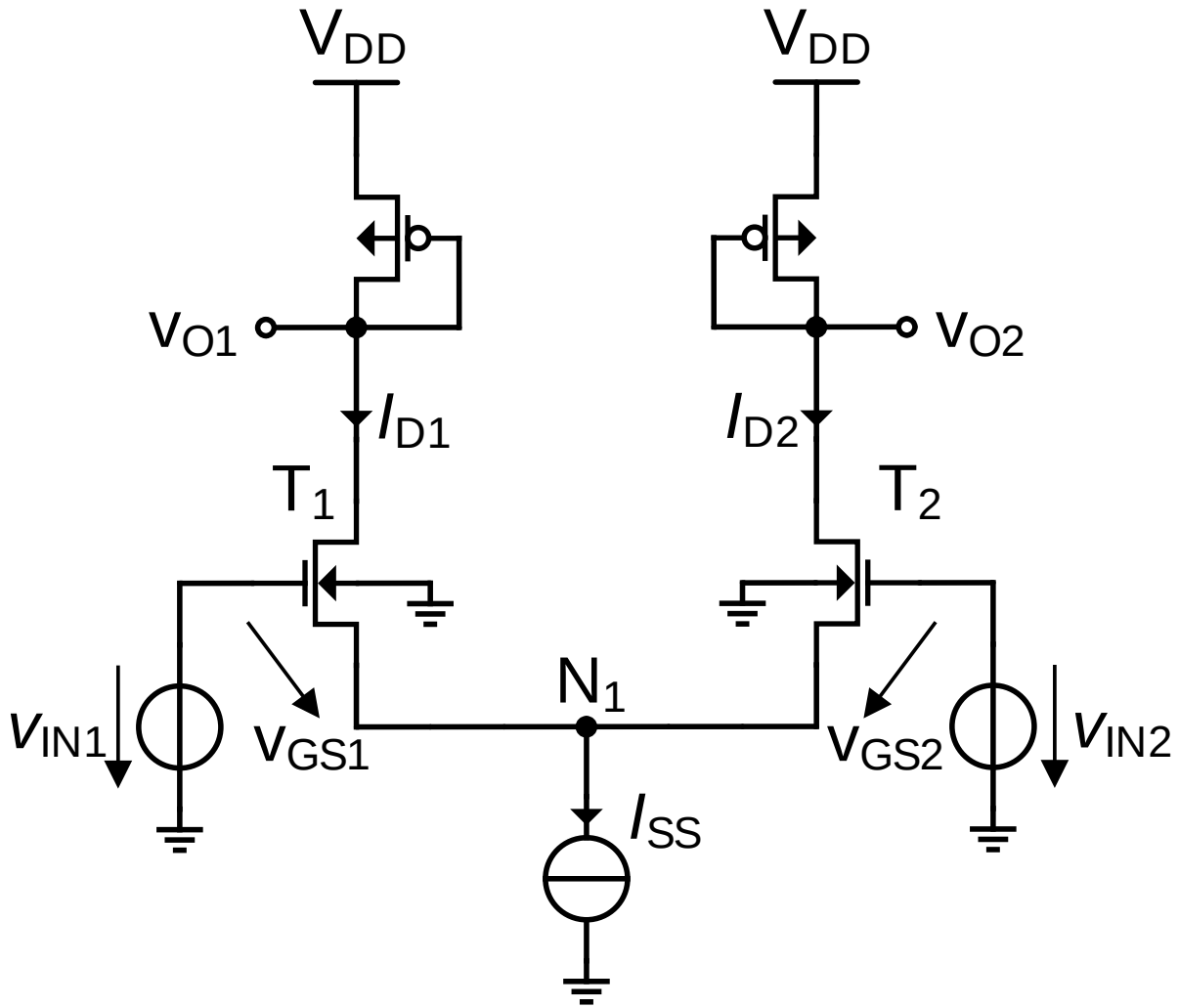
mismatch between current source (load) & current sink (biasing)

no feedback → $v_{O1,2}$ @ $V_{SS}$ or $V_{DD}$

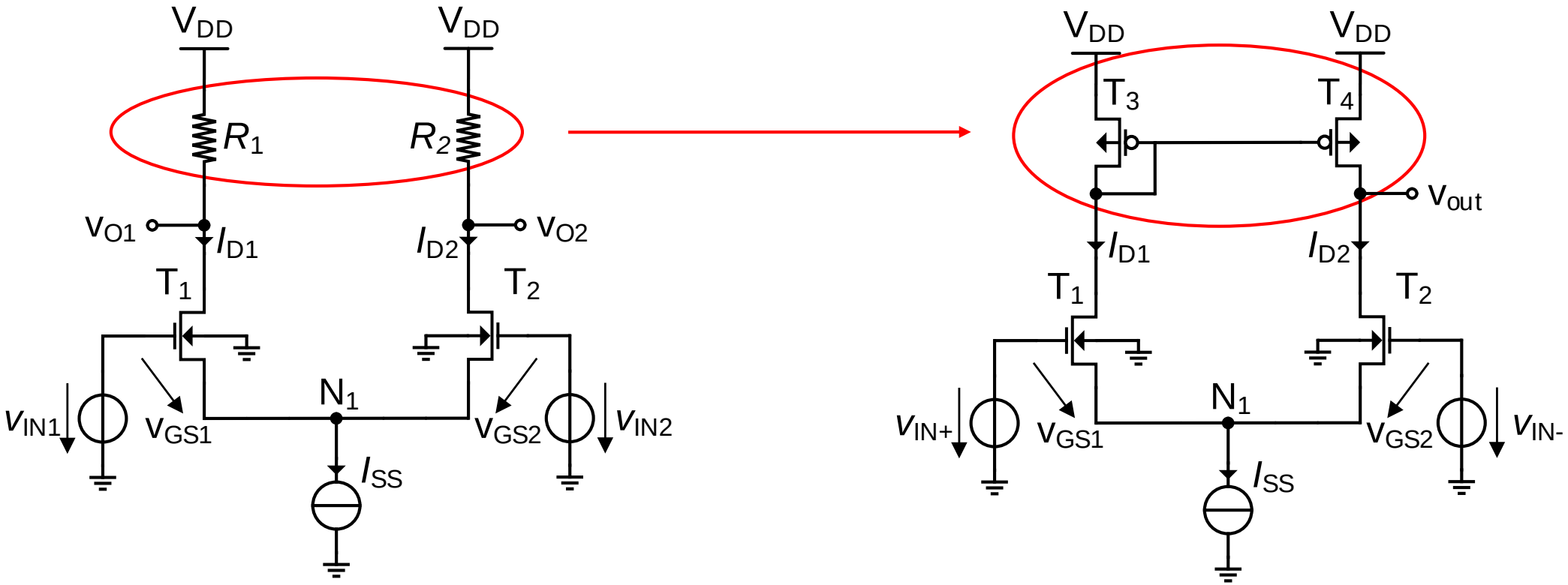
# MOS-Diode Load

advantage:
self-biasing of the load elements
$V_{GS3,4} = f(I_{SS})$

disadvantage:
low gain



# Current Mirror Load




differential input

differential output

differential input

single-ended output

# Current Mirror Load – DM Small-Signal Amplification (Voltage)

Apply the Principle of Superposition to determine the small-signal output voltage $v_{\text{out}}$, i.e.,

$$v_{\text{out}} = v_{\text{out+}} + v_{\text{out-}} = A_{\text{V+}} v_{\text{in+}} + A_{\text{V-}} v_{\text{in-}}$$

with

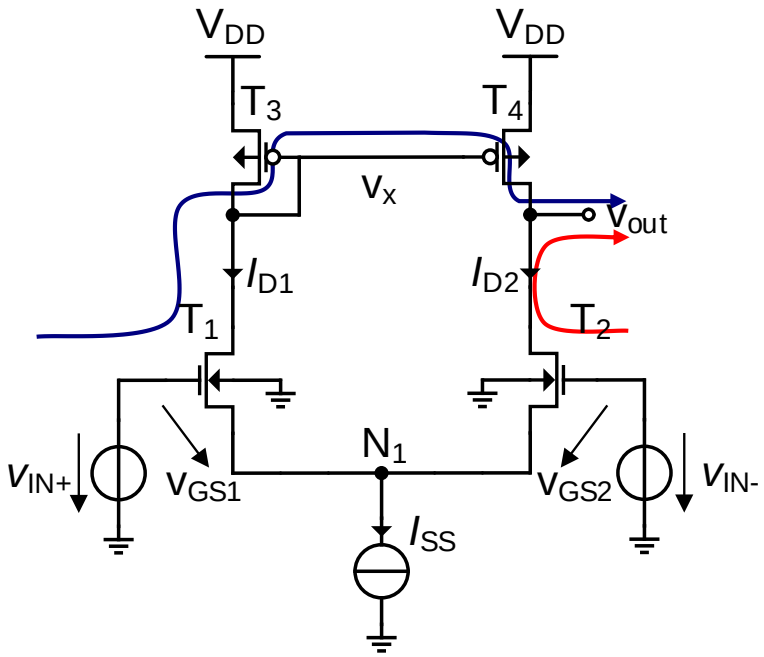
$$v_{\text{in+}} = \frac{v_{\text{id}}}{2} \text{ and } v_{\text{in-}} = -\frac{v_{\text{id}}}{2}$$

$$A_{\text{V-}} = \frac{v_{\text{out-}}}{v_{\text{in-}}} \approx -g_{\text{m2}}(r_{\text{ds2}} || r_{\text{ds4}})$$

$$A_{\text{V+}} = \frac{v_{\text{out+}}}{v_{\text{in+}}} \approx \frac{v_{\text{x}}}{v_{\text{in+}}} \frac{v_{\text{out+}}}{v_{\text{x}}} = -\frac{g_{m1}}{\cancel{g_{m3}}} \left(-\cancel{g_{m4}}(r_{\text{ds2}} || r_{\text{ds4}})\right)$$

$$v_{\text{out}} = A_{\text{V+}} v_{\text{in+}} + A_{\text{V-}} v_{\text{in-}} \approx g_{m1}(r_{\text{ds2}} || r_{\text{ds4}})(v_{\text{in+}} - v_{\text{in-}})$$

$$\frac{v_{\text{out}}}{v_{\text{in+}} - v_{\text{in-}}} = \frac{v_{\text{out}}}{v_{\text{id}}} \approx g_{\text{m1,2}}(r_{\text{ds2}} || r_{\text{ds4}})$$




Similar DM small-signal gain as in case of a differential output (current source load)!

# Current Mirror Load – DM Small-Signal Amplification (Current)

$$A_V = \frac{v_{out}}{v_{in}} = -G_m r_{out} = -\frac{i_{out}}{v_{in}} r_{out}$$

$$i_1 = g_{m1}\left(\frac{v_{in}}{2}\right)$$

$$i_2 = g_{m2}\left(-\frac{v_{in}}{2}\right)$$

$$i_{out} = i_4 - i_2 = g_{m1}\frac{v_{in}}{2} + g_{m2}\frac{v_{in}}{2}$$

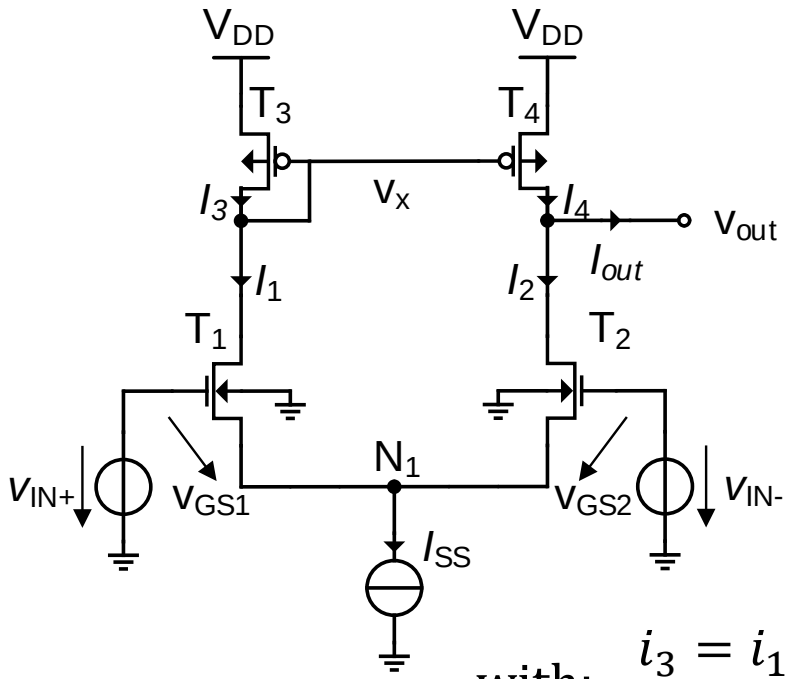
$$= (g_{m1} + g_{m2})\frac{v_{in}}{2}$$

$$\Rightarrow i_{out} = g_{m1,2} v_{in}$$

$$A_V = -g_{m1,2}\left(r_{ds2} \| r_{ds4}\right)$$

$$r_{out} = r_{ds2} \| r_{ds4}$$

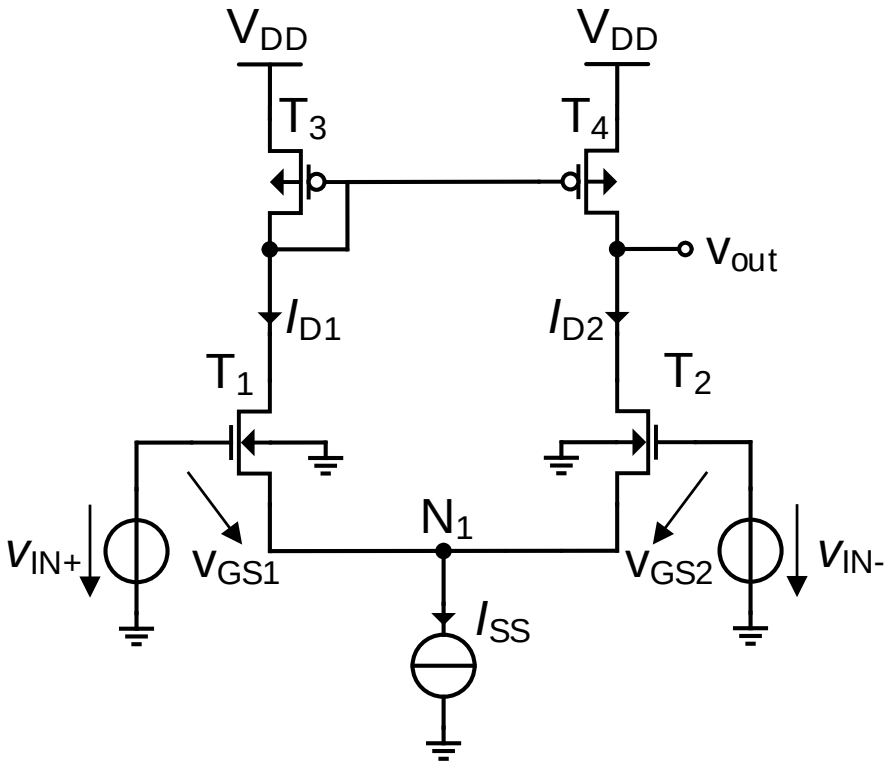
$$i_{out} = ?$$




with: $\left.\begin{matrix} i_3 = i_1 \\ i_4 = i_3 \end{matrix}\right\} i_4 = i_1$

# Current Mirror Load – Summary Characteristics

- self-biasing of all transistors by current source $I_{SS}$
- differential input to single-ended output conversion
- high gain: $A_V = -g_{m1,2}\left(r_{ds2} \| r_{ds4}\right)$
- higher output swing compared with “MOS diode” load, i.e., $V_{DD} - V_{DSsat}$ versus $V_{DD} - V_{th}$



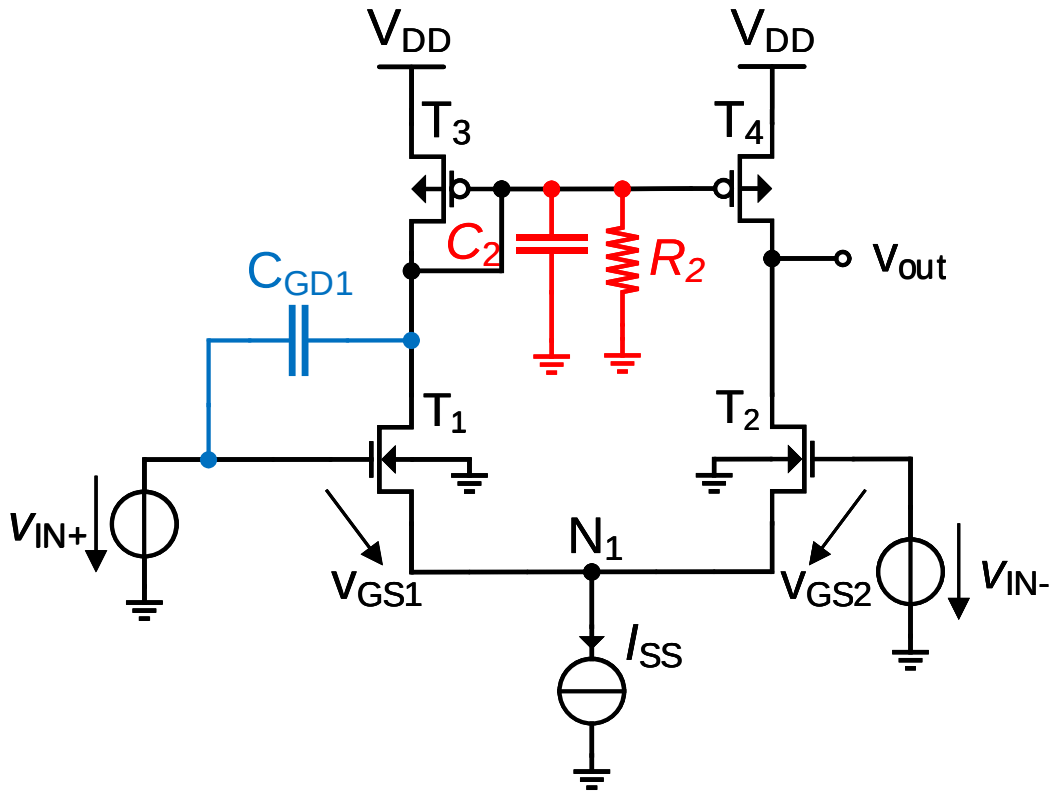
# Overview

- Motivation
- Differential Stage
- **Frequency Response**

# Determination of Zeros & Poles

general notes:

- A pole is given at every node in the signal path.
- It is determine by the overall capacitor from the considered node to “small-signal AC ground” and the small-signal resistance connected parallel, e.g., $-1/r_2C_2$.
- A zero is given by any amplifying stage in the signal path.
- It is determined by the gate-drain capacitor $C_{\mathrm{GD}}$ and the transconductance $g_{\mathrm{m}}$ of the input transistor of the amplifying stage, e.g., $+g_{m1}/C_{\mathrm{GD1}}$.



# Determination Poles I - Capacitors

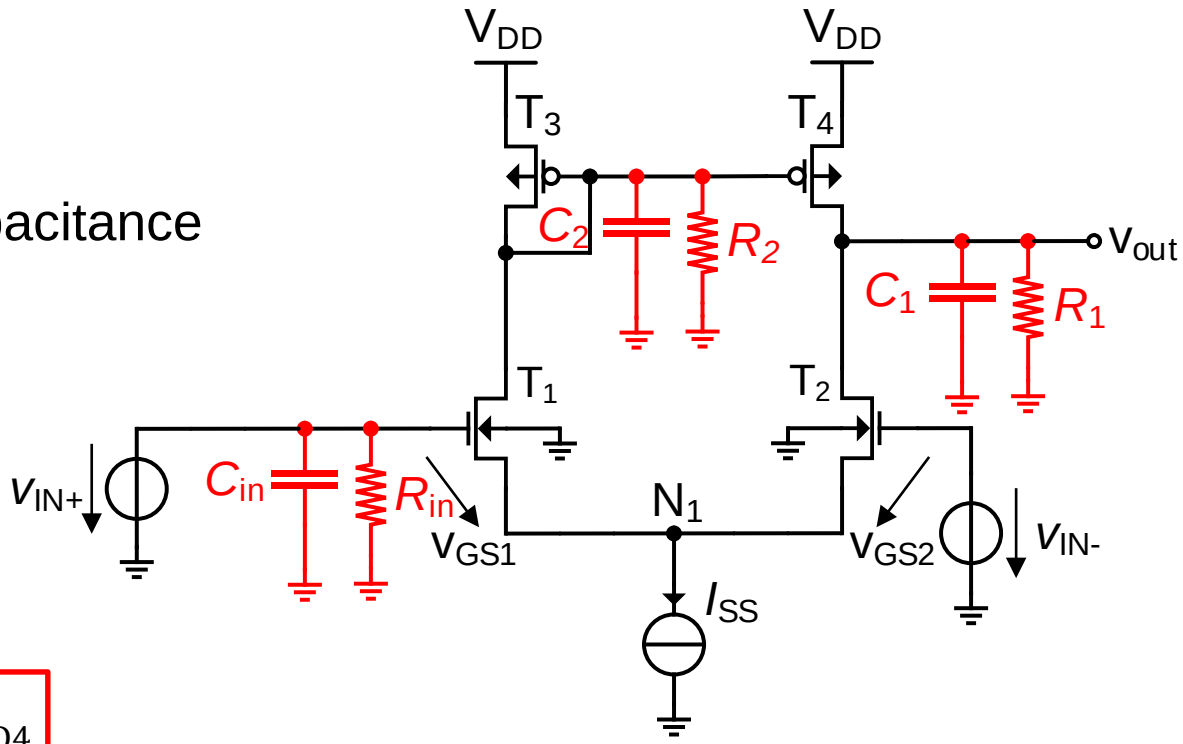
general approach:

- Apply Miller-Theorem to determine the overall capacitance from a node to ground.

$$C_{\text{in}} \approx C_{\text{GS1}} + \frac{g_{\text{m1}}}{g_{\text{m2}}} C_{\text{GD1}}$$

$$C_1 \approx C_{\text{L}} + C_{\text{DB2}} + C_{\text{DB4}} + C_{\text{GD2}} + C_{\text{GD4}}$$

$$C_2 \approx C_{\text{GD1}} + C_{\text{GS3}} + C_{\text{GS4}} + C_{\text{DB3}} + g_{\text{m4}}(r_{\text{ds2}} || r_{\text{ds4}})\, C_{\text{GD4}}$$




Miller-Capacitor at the input / output of an amplifying stage

- However, the Miller-amplification is to be evaluated at the resulting pole frequency (which is not known a priori) and not at DC (as typically done in hand calculations)!
  → Assumption of Miller-effect may be false! → To be verified subsequently!

# Determination Poles II – Resistors

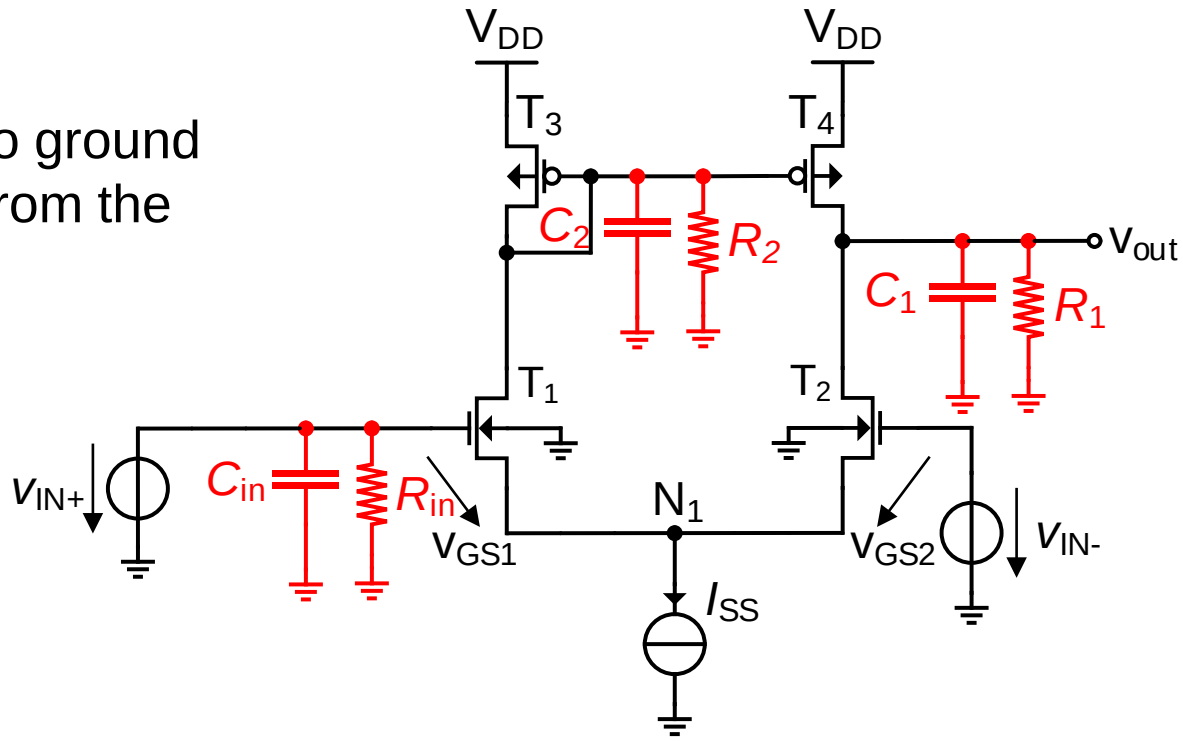
general approach:

- Determine the overall resistance from a node to ground by evaluating the parallel resistance resulting from the resistances seen “left” and “right” of the node.

$$r_{\text{in}} \approx r_{\text{sig}} || \infty \approx r_{\text{sig}}$$

$$r_1 \approx (r_{\text{ds2}} || r_{\text{ds4}}) || \infty \approx r_{\text{ds2}} || r_{\text{ds4}}$$

$$r_2 \approx \left( r_{\text{ds1}} || \frac{1}{g_{\text{m3}}} \right) || \infty \approx \frac{1}{g_{m3}}$$

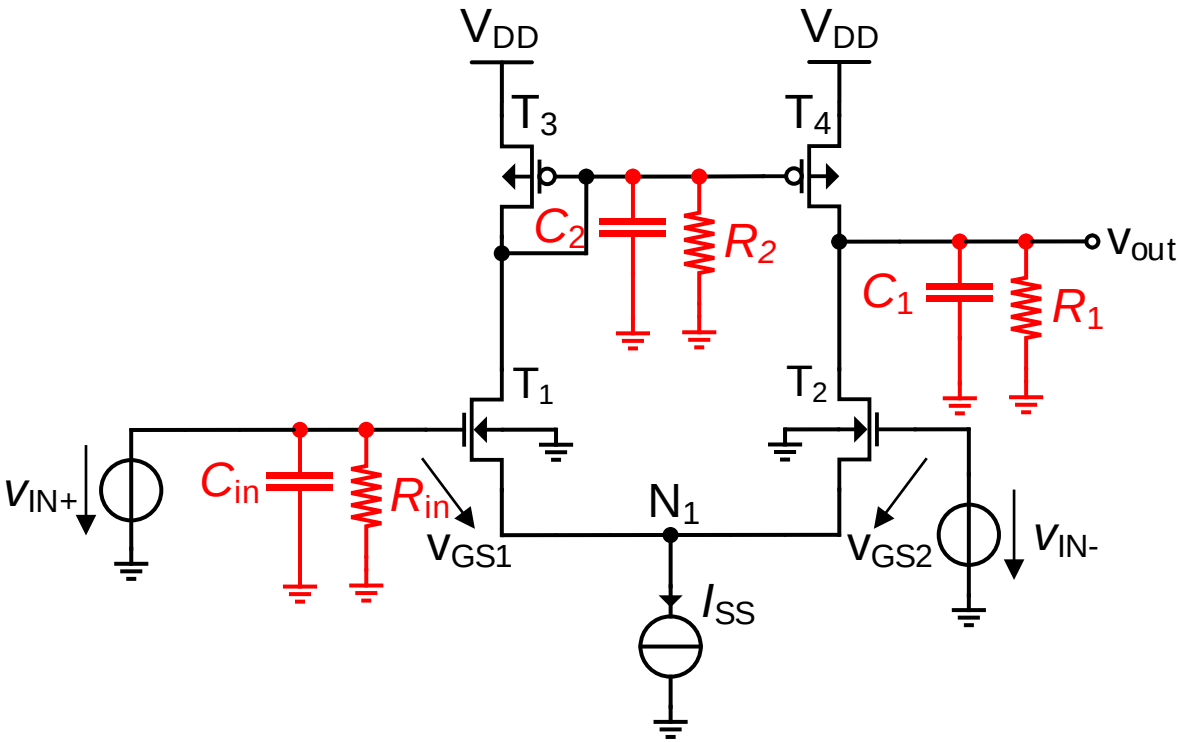


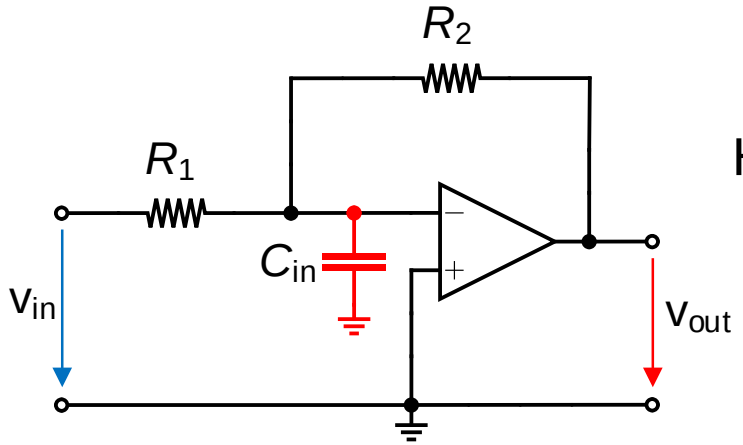
# Determination Poles – Input Pole

$$r_{\text{in}} \approx r_{\text{sig}} || \infty \approx r_{\text{sig}}$$

$$C_{\text{in}} \approx C_{\text{GS1}} + \frac{g_{\text{m1}}}{g_{\text{m2}}} C_{\text{GD1}}$$

We typically do not consider the pole at the input since the resistance $r_{\text{sig}}$ of an ideal voltage source is $0\ \Omega$, i.e., the pole is located at infinity.




However, it is to be considered in feedback systems (no topic of this lecture).

# Transfer Function $v_{\mathrm{out}} / v_{\mathrm{d}}$ – Derivation I

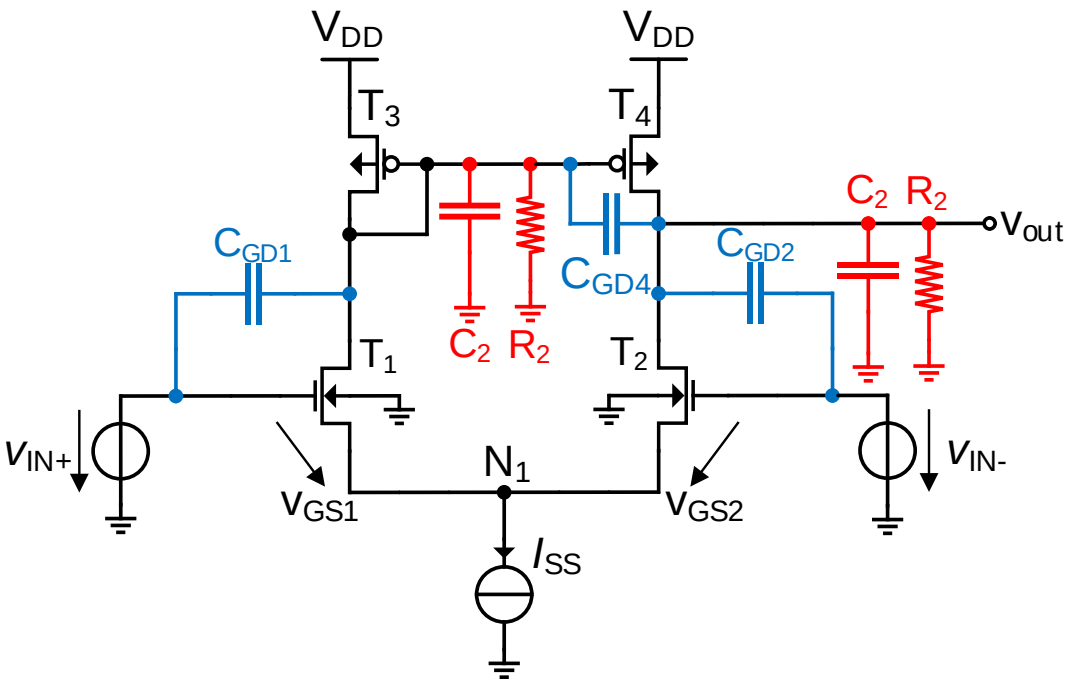
Neglecting the pole at the input and noticing that the (infinite) input resistance of a subsequent stage has no impact on the output resistance of the previous stage and thus on the overall resistance of a node to ground, i.e.,

$$r_1 \approx (r_{\mathrm{ds2}} || r_{\mathrm{ds4}}) || \infty \approx r_{\mathrm{ds2}} || r_{\mathrm{ds4}}$$

$$r_2 \approx \left( r_{\mathrm{ds1}} || \frac{1}{g_{\mathrm{m3}}} \right) || \infty \approx \frac{1}{g_{m3}}$$

we can derive the transfer functions from $v_{\mathrm{in+}}$ and $v_{\mathrm{in-}}$ to $v_{\mathrm{out}}$ by multiplying the (complex) transfer functions of single-stage amplifiers.

$$\underline{A}_{\mathrm{V}} = -\frac{g_{\mathrm{m}} - sC_{\mathrm{GD}}}{g_{\mathrm{out}} + sC_{\mathrm{out}}}$$ (see chapter 5)

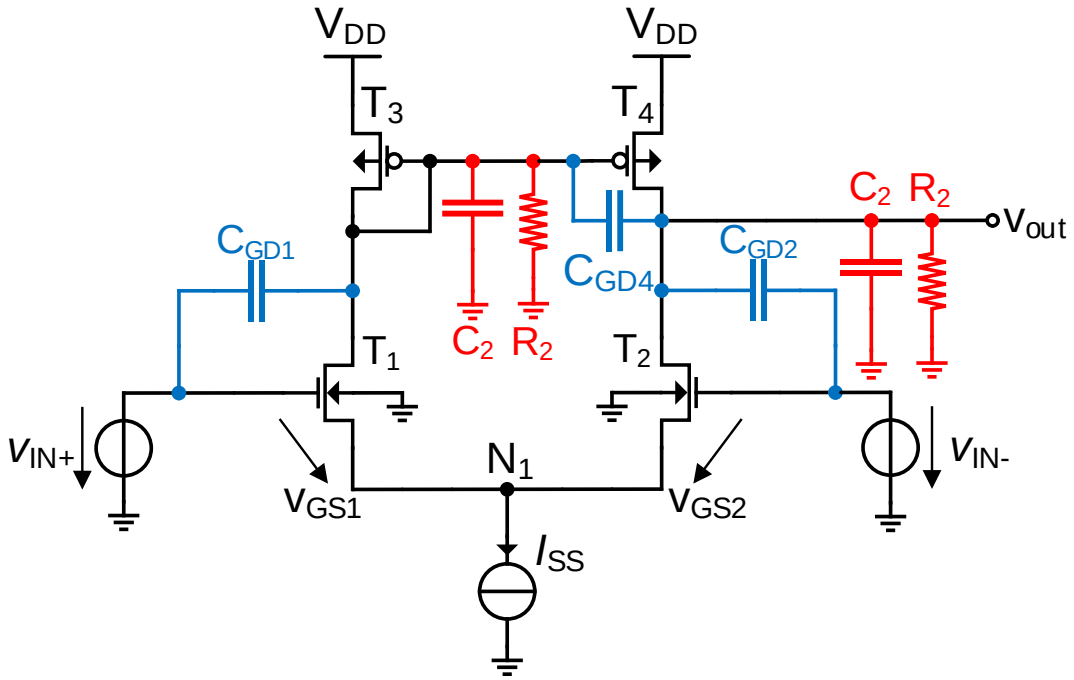


# Transfer Function $v_{out} / v_d$ – Derivation II

We thus get

$$A_{v-} = \frac{\underline{v}_{out}}{\underline{v}_{in-}} = -\frac{g_{m2} - sC_{GD2}}{g_1 + sC_1}$$

$$A_{v+} = \frac{\underline{v}_{out}}{\underline{v}_{in+}} = \left(-\frac{g_{m1} - sC_{GD1}}{g_2 + sC_2}\right)\left(-\frac{g_{m4} - sC_{GD4}}{g_1 + sC_1}\right)$$




The transfer function $v_{out} / v_d$ is found by applying the Principle of Superposition

$$A_{v+}\underline{v}_{in+} + A_{v-}\underline{v}_{in-} = A_{v+}\frac{v_d}{2} + A_{v-}\left(-\frac{v_d}{2}\right) \rightarrow A_v = \frac{v_{out}}{v_d} = \frac{A_{v+} - A_{v-}}{2}$$

# Transfer Function $v_{\mathrm{out}} / v_{\mathrm{d}}$ – Derivation III

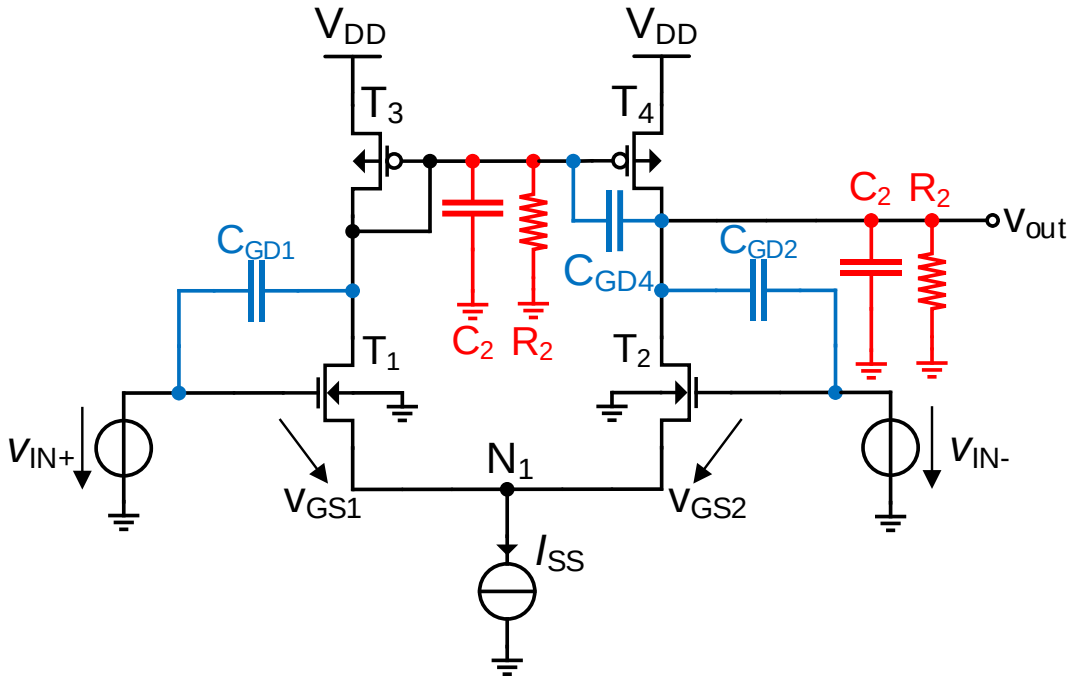
Due to symmetry, i.e., $T_1 = T_2$ and $T_3 = T_4$, we get

$$A_{\mathrm{v}} = \frac{v_{\mathrm{out}}}{v_{\mathrm{d}}} = \frac{A_{\mathrm{v+}} - A_{\mathrm{v-}}}{2}$$

$$= \frac{1}{2}\frac{g_{\mathrm{m1,2}} - sC_{\mathrm{GD1,2}}}{g_1 + sC_1}\left(\frac{g_{\mathrm{m4}} - sC_{\mathrm{GD4}}}{g_2 + sC_2} + 1\right)$$

$$\approx \frac{1}{2}\frac{g_{\mathrm{m1,2}} - sC_{\mathrm{GD1,2}}}{g_1 + sC_1}\left(\frac{2g_{\mathrm{m3,4}} + sC_2}{g_{m3,4} + sC_2}\right)$$

$$\boxed{A_{\mathrm{v}} = \frac{1}{2}\frac{(g_{\mathrm{m1,2}} - sC_{\mathrm{GD1,2}})(r_{\mathrm{ds2}} || r_{\mathrm{ds4}})}{1 + sC_1(r_{\mathrm{ds2}} || r_{\mathrm{ds4}})}\left(\frac{2g_{\mathrm{m3,4}} + sC_2}{g_{m3,4} + sC_2}\right)}$$



# Transfer Function $v_{\text{out}} / v_{\text{d}}$

+20dB/decade
-90°

+20dB/decade
+90°

$$\text{zero } \omega_{z2} \approx \frac{g_{m1,2}}{C_{GD1,2}} \qquad \text{zero } \omega_{z1} \approx -\frac{2g_{m3,4}}{C_2}$$

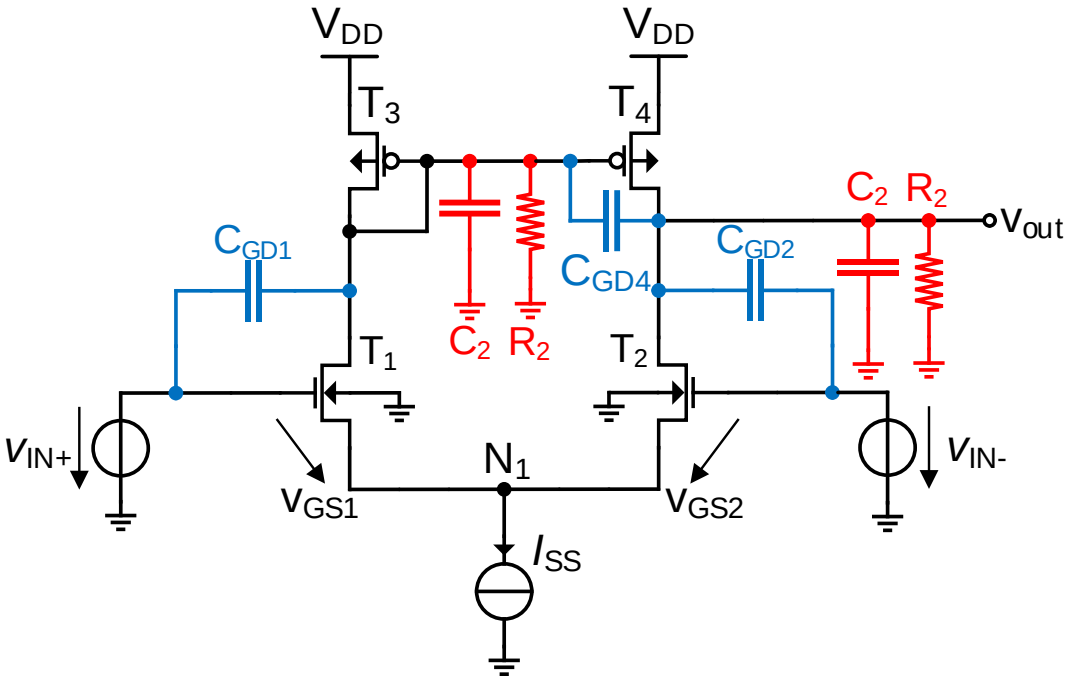
$$A_v = \frac{1}{2} \frac{(g_{m1,2} - sC_{GD1,2})(r_{ds2} || r_{ds4})}{1 + sC_1(r_{ds2} || r_{ds4})} \left( \frac{2g_{m3,4} + sC_2}{g_{m3,4} + sC_2} \right)$$

dominant pole
(output pole)

second pole
(mirror pole)

$$\omega_{p1} \approx -\frac{1}{C_1(r_{ds2} || r_{ds4})} \qquad \omega_{p2} \approx -\frac{g_{m3,4}}{C_2}$$

-20dB/decade
-90°



# Transfer Function $v_{\mathrm{out}} / v_{\mathrm{d}}$ – Notes

$$A_{\mathrm{v}} = \frac{1}{2}\frac{(g_{\mathrm{m1,2}} - sC_{\mathrm{GD1,2}})(r_{\mathrm{ds2}} || r_{\mathrm{ds4}})}{1 + sC_1(r_{\mathrm{ds2}} || r_{\mathrm{ds4}})}\left(\frac{2g_{\mathrm{m3,4}} + sC_2}{g_{m3,4} + sC_2}\right)$$

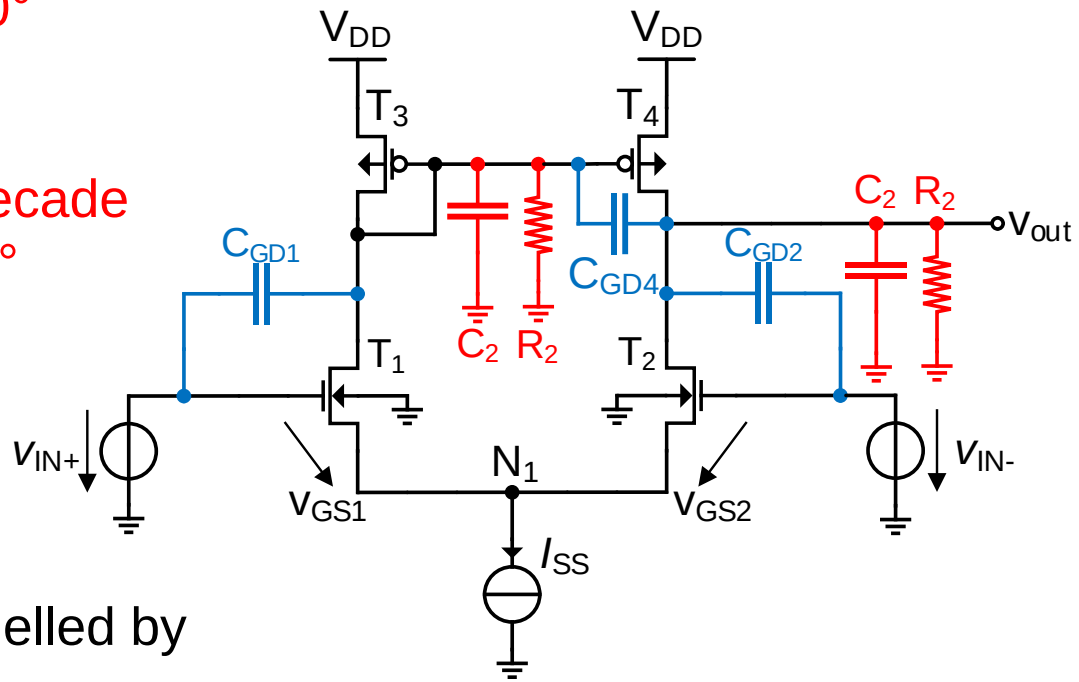
+20dB/decade
+90°

-20dB/decade
-90°

- small-signal DC gain (s = 0)

  $A_{\mathrm{v}} = g_{\mathrm{m1,2}}(r_{\mathrm{ds2}} || r_{\mathrm{ds4}})$

- @ current mirror node:
  zero frequency is twice the pole frequency
  → pole-zero compensation
  → differential stage with current mirror load can be modelled by




$$\omega_{\mathrm{p1}} \approx -\frac{1}{C_1(r_{\mathrm{ds2}} || r_{\mathrm{ds4}})} \text{ and } \omega_{\mathrm{z2}} \approx \frac{g_{\mathrm{m1,2}}}{C_{\mathrm{GD1,2}}}$$

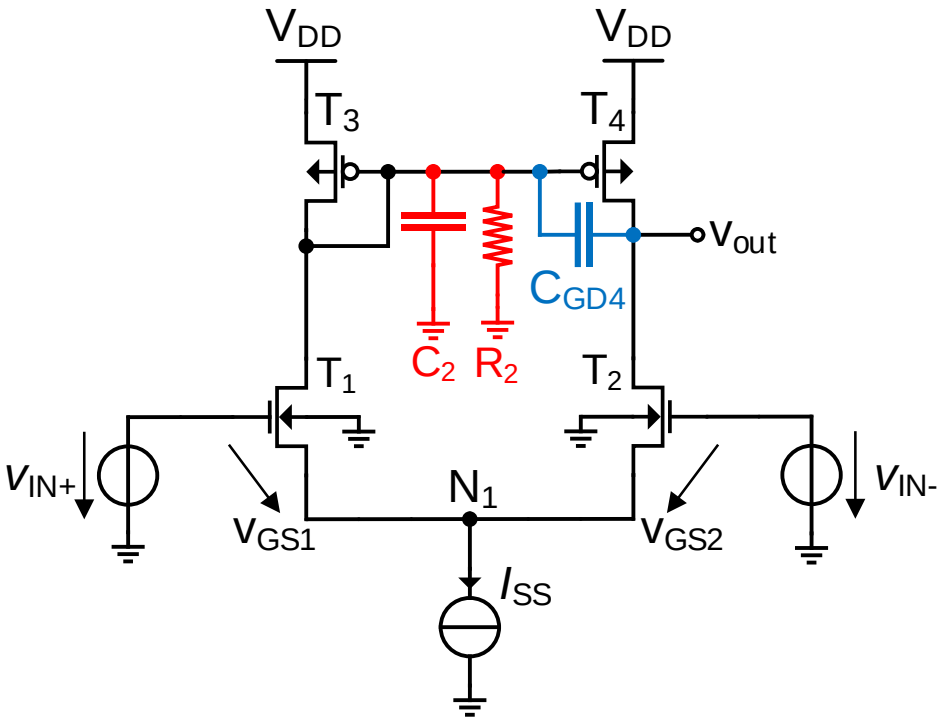
However, the zero is at a much higher frequency than the pole since $g_{\mathrm{m}} \gg g_{\mathrm{ds}}$.

# Verification Miller-Amplification

$$A_v = \frac{1}{2}\frac{(g_{m1,2} - sC_{GD1,2})(r_{ds2}||r_{ds4})}{1 + sC_1(r_{ds2}||r_{ds4})}\left(\frac{2g_{m3,4} + sC_2}{g_{m3,4} + sC_2}\right)$$

$$C_2 \approx C_{GD1} + g_{m4}(r_{DS4}||r_{DS2})C_{GD4} + C_{GS3} + C_{GS4} + C_{DB1} + C_{DB3}$$
$$\approx g_{m4}(r_{DS4}||r_{DS2})C_{GD4} + C_{GS3} + C_{GS4}$$

The Miller-amplification is to be evaluated at the pole frequency $\omega_{p2}$.
However, the frequency behavior of the Miller-amplification is determined by $\omega_{p1}$.




If $\omega_{p2} \ll \omega_{p1} \rightarrow A_{MILLER}(\omega_{p2}) \approx A_{MILLER}(\omega = 0) = g_{m4}(r_{ds4}||r_{ds2})$

Miller capacitor dominant!

$$\rightarrow \omega_{p2} \approx \frac{g_{m3,4}}{g_{m4}(r_{ds4}||r_{ds2})C_{GD4}}$$

If $\omega_{p2} \gg \omega_{p1} \rightarrow A_{MILLER}(\omega_{p2})$ small

Miller capacitor not dominant!

$$\rightarrow \omega_{p2} \approx \frac{g_{m3,4}}{2C_{GS3,4}}$$

# Frequency Response – Example

$$C_1 \approx C_\mathrm{L} = 100\ \mathrm{fF}$$
$$C_2 \approx 2C_{\mathrm{GS3,4}} + \cancel{A_\mathrm{Miller}(\omega_\mathrm{p2})C_\mathrm{GD4}} = 150\ \mathrm{fF}$$

Negligible!
$A_\mathrm{MILLER}(\omega_\mathrm{p2})$ is small!

dominant pole (output)

$$\omega_\mathrm{p1} = 2\pi f_\mathrm{p1} \approx \frac{1}{C_1(r_\mathrm{ds2}||r_\mathrm{ds4})} \Rightarrow f_\mathrm{p1} \approx 3.54\ \mathrm{MHz}$$

zero (input)

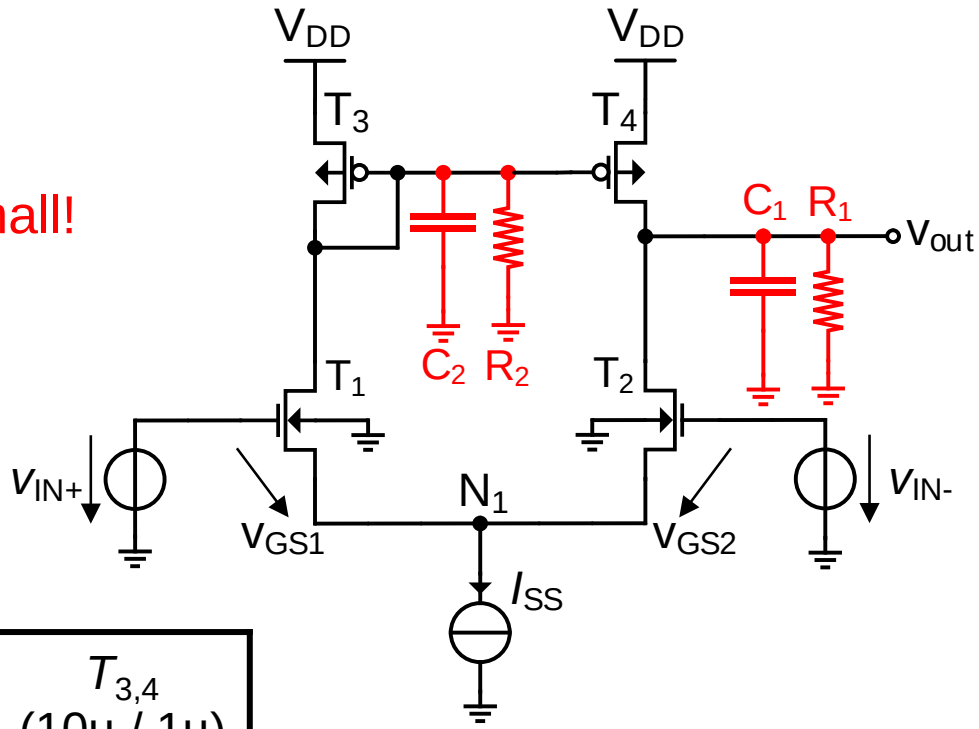
$$\omega_{z2} = 2\pi f_{z2} \approx \frac{g_\mathrm{m1,2}}{C_\mathrm{GD1,2}} \Rightarrow f_{z2} \approx 18\ \mathrm{GHz}$$

second pole (current mirror load)

$$\omega_\mathrm{p2} = 2\pi f_\mathrm{p2} \approx \frac{g_\mathrm{m3,4}}{C_2} \Rightarrow f_\mathrm{p2} \approx 125.2\ \mathrm{MHz}$$

zero (current mirror load)

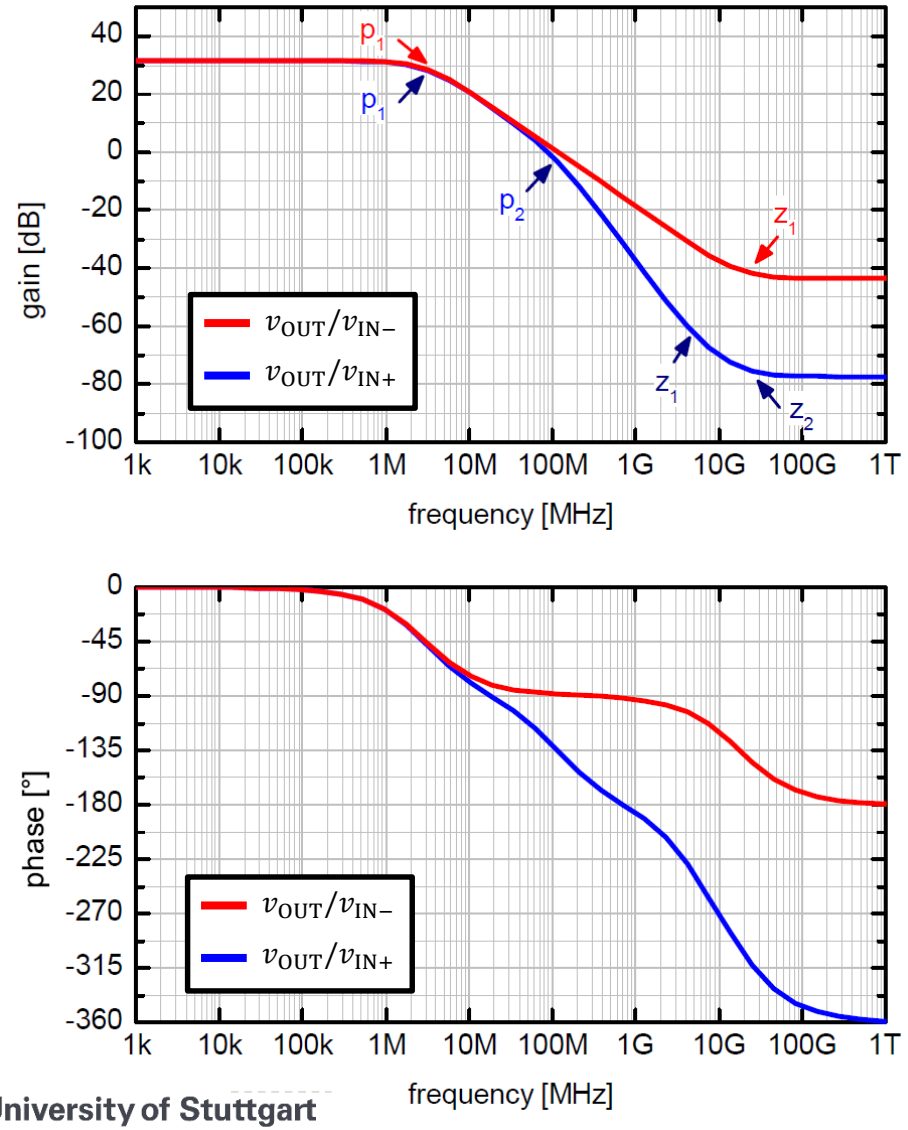
$$\omega_{z1} = 2\pi f_{z1} \approx \frac{2g_\mathrm{m3,4}}{C_2} \Rightarrow f_{z1} \approx 250.4\ \mathrm{MHz}$$

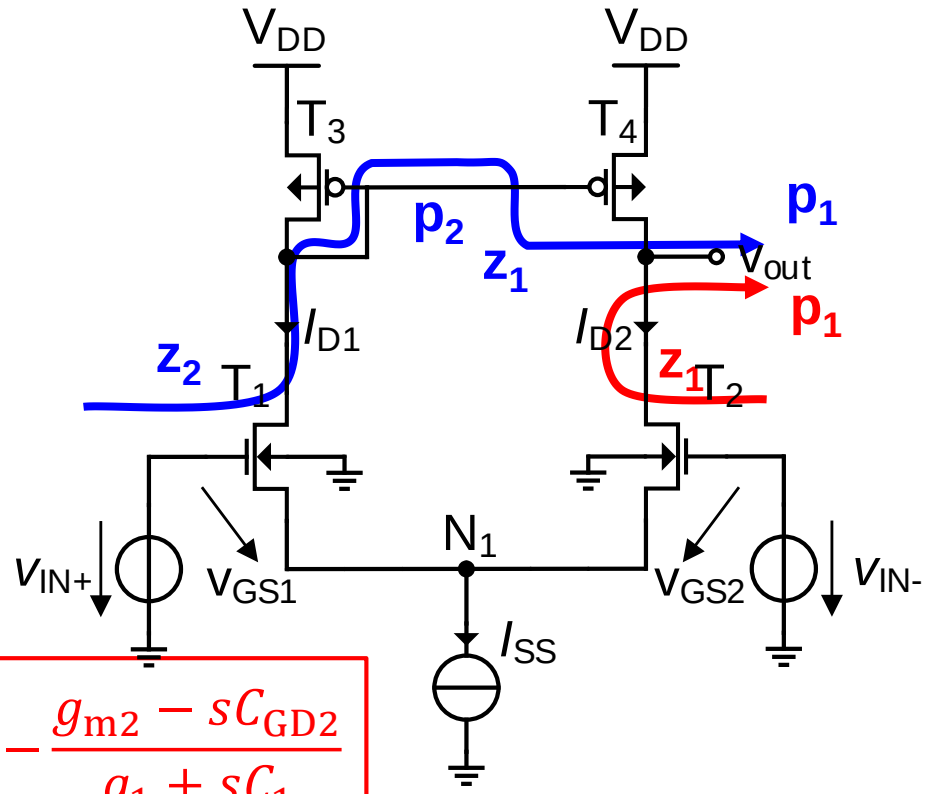



$I_\mathrm{SS} = 20\ \mu\mathrm{A}$

| | $T_{1,2}$ (5µ / 1µ) | $T_{3,4}$ (10µ / 1µ) |
|---|---|---|
| $C_\mathrm{DB}$ | 2.1 fF | 7.4 fF |
| $C_\mathrm{GD}$ | 1.5 fF | 3.2 fF |
| $C_\mathrm{GS}$ | 32.2 fF | 73.3 fF |
| $g_\mathrm{m}$ | 170 µS | 118 µS |
| $g_\mathrm{ds}$ | 1.7 µS | 526 nS |

# (Simulated) Bode Plot – $v_{out} / v_{in+}$ & $v_{out} / v_{in-}$

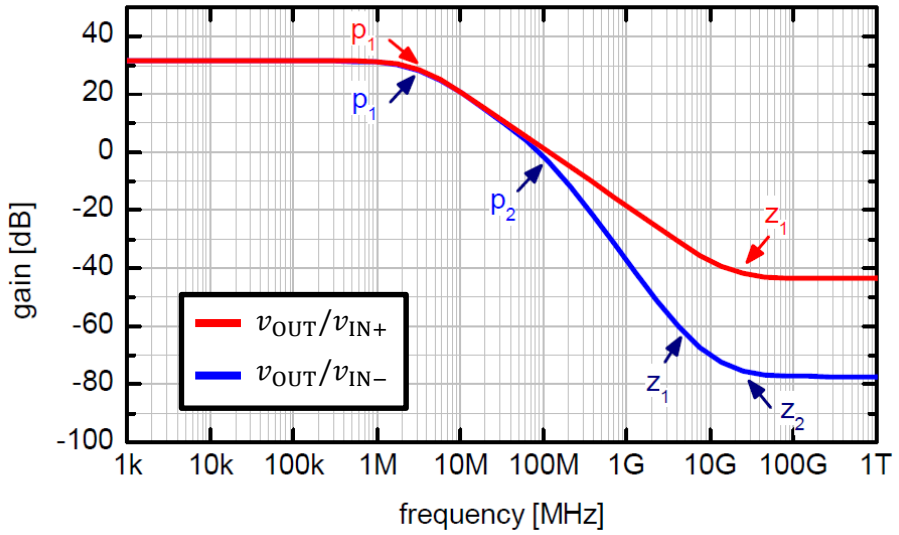


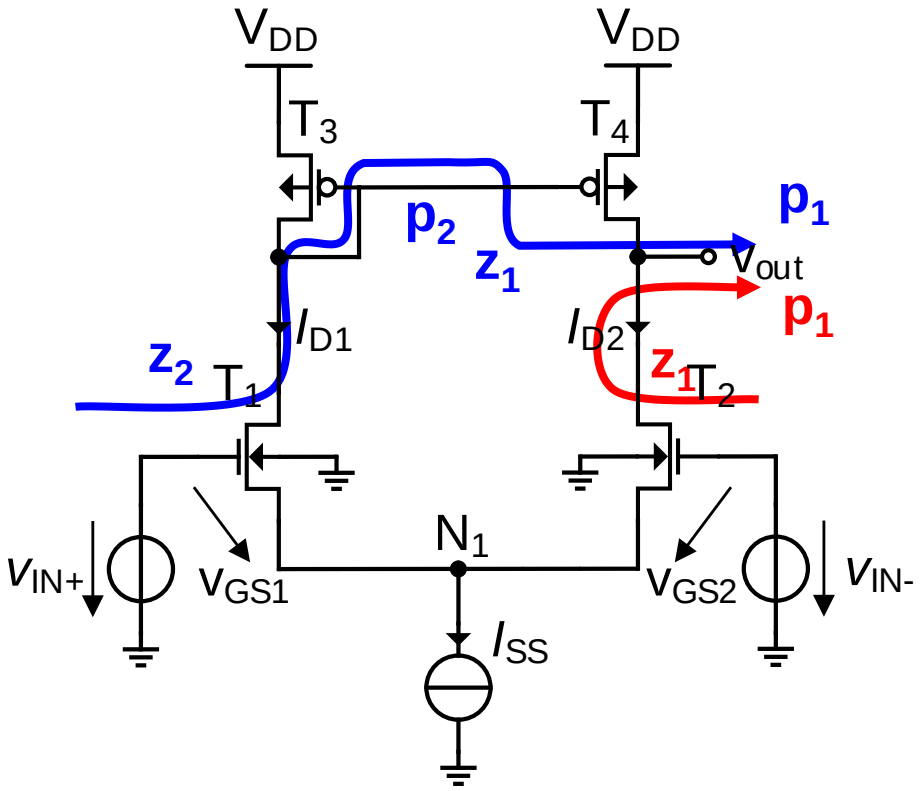




$$A_{v-} = \frac{v_{out}}{v_{in-}} = -\frac{g_{m2} - sC_{GD2}}{g_1 + sC_1}$$

$$A_{v+} = \frac{v_{out}}{v_{in+}} = \left(-\frac{g_{m1} - sC_{GD1}}{g_2 + sC_2}\right)\left(-\frac{g_{m4} - sC_{GD4}}{g_1 + sC_1}\right)$$

# Comparison – Calculated vs. Simulated Pole/Zero Frequencies




$$f_{p1} \approx \frac{1}{2\pi} \frac{1}{C_1 (r_{ds2} \| r_{ds4})} = 2.4\ \text{MHz} \qquad f_{z1} \approx \frac{1}{2\pi} \frac{g_{m4}}{C_{GD2}} = 18\ \text{GHz}$$

$$f_{p1} \approx \frac{1}{2\pi} \frac{1}{C_1 (r_{ds2} \| r_{ds4})} = 2.4\ \text{MHz} \qquad f_{z1} \approx \frac{1}{2\pi} \frac{g_{m4}}{C_{GD4}} = 5.9\ \text{GHz}$$

$$f_{p2} \approx \frac{1}{2\pi} \frac{g_{m3}}{2C_{GS3,4}} = 125.2\ \text{MHz} \qquad f_{z2} \approx \frac{1}{2\pi} \frac{g_{m1}}{C_{GD1}} = 18\ \text{GHz}$$

# Bode Plot – Differential Stage

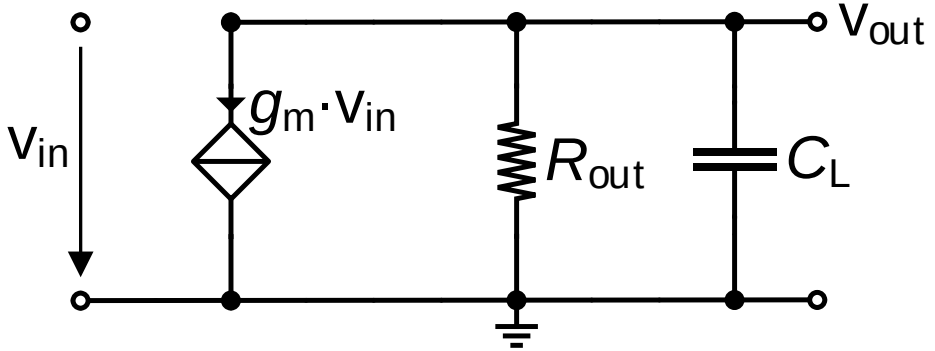
$$A_v = \frac{1}{2}\frac{(g_{m1,2} - sC_{GD1,2})(r_{ds2}||r_{ds4})}{1 + sC_1(r_{ds2}||r_{ds4})}\left(\frac{2g_{m3,4} + sC_2}{g_{m3,4} + sC_2}\right)$$

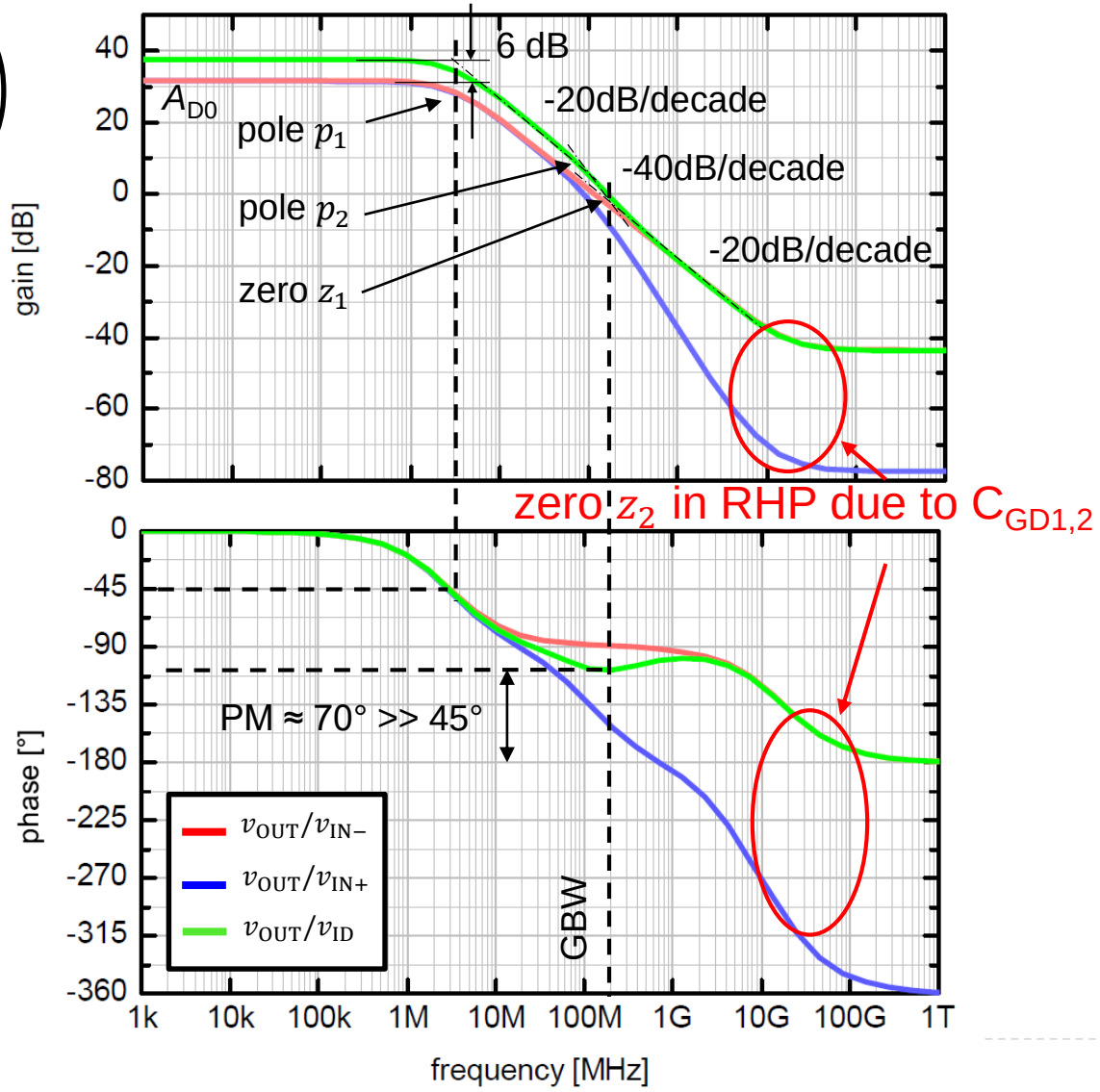
$f_{p1} \approx 2.4$ MHz (output)

$f_{p2} \approx 125.2$ MHz (current mirror)

$f_{z1} \approx 250.4$ MHz (current mirror)

$f_{z2} \approx 18$ GHz (input)



# Evaluation

https://evasysw.uni-stuttgart.de/evasys/online.php?p=ZPEU6

